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1. Introduction

Our final year project revolves around an Android application designed to harness the power of artificial intelligence (AI) and machine learning (ML) for analyzing cricket bowling videos recorded with a regular camera. With the growing interest in cricket and the desire for players to enhance their skills, we aim to create a platform that delivers valuable insights and constructive feedback based on their bowling performances. This document will detail our project vision, software requirements, and include UML designs and ER diagrams to facilitate the development process.

2. Project Vision

2.1. Problem Statement

Many amateur and semi-professional cricketers find it challenging to self-assess and improve their techniques due to the absence of immediate, actionable feedback on their bowling styles. Conventional coaching methods can be lengthy and often lack the detailed analysis players need. Our application seeks to fill this void by leveraging AI and ML to analyze bowling videos, pinpoint errors, and generate personalized feedback, ultimately enhancing training quality.

2.2. Business Opportunity

As the need for effective training solutions in cricket rises, particularly among players looking to refine their skills independently, our user-friendly application can meet this demand by analyzing bowling techniques. We also see potential partnerships with cricket academies and online coaching platforms, which could create additional revenue through app sales and subscription models.

2.3. Objectives

Our main goal is to develop a platform that allows users to upload their bowling videos and receive thorough analyses and feedback. By utilizing AI and ML technologies, we aim to provide insights that empower players to improve their performance, particularly benefiting those in Pakistan where access to professional coaching can be limited.

2.4. Project Scope

With cricket's increasing popularity, the demand for effective training tools is evident. Our application will enable players to record their bowling videos, which will then be processed using sophisticated algorithms to identify technical flaws and offer constructive feedback. This platform will serve as a comprehensive training tool, helping users track their progress over time.

2.5. Constraints

Our project faces several constraints, including scope, timeline, and budget. We must adhere to defined project goals and deliverables, with specific deadlines for both FYP 1 and FYP 2 that set a timeline for

our development. The team consists of only three developers, which may impact the speed of progress. Additionally, access to video processing technology and the necessity for user-friendly interfaces will inform our design choices.

2.6. Stakeholder Description

The primary stakeholders in this project include:

- **Developers:** Internal stakeholders tasked with the design and implementation of the application, focused on delivering a functional and efficient product.
- **Users:** External stakeholders, mainly cricket players and coaches, who will use the application to enhance their skills. Their satisfaction and experience will directly influence the application's success.

2.6.1. Stakeholder Summary

- **Stakeholder 1: Developers**
 - Role: Ensure the application meets functional requirements and is user-friendly.
- **Stakeholder 2: Users**
 - Role: Seek an intuitive experience that enhances their training and performance.

2.6.2. Key High-Level Goals and Problems of Stakeholders

High-Level Goals:

1. Enhance user experience with actionable insights.
2. Ensure timely application deployment.
3. Provide accurate and valuable analyses.
4. Fulfill all specified product requirements.
5. Maintain ongoing system updates and maintenance.

Problems of Stakeholders:

1. Challenges in self-assessment and improvement.
2. Difficulties in effectively testing the application.
3. Ensuring accessibility across different devices.
4. Engaging users to consistently use the app for their training.
5. Adapting to AI-driven feedback.

3. SRS

The “Khiladi” app offers a platform for cricket players to upload their bowling videos, which are analyzed using AI and machine learning. Users can create accounts, personalize their profiles, and submit videos for analysis. After processing, the app generates feedback and identifies areas for improvement,

helping users enhance their bowling techniques. Users can also track their progress and receive personalized training plans based on their analyses.

3.1. List of Features

- User account creation and management.
- Profile customization (age, bowling style, experience).
- Video upload functionality.
- AI-driven video analysis to identify mistakes.
- Personalized feedback generation.
- Progress tracking over time.
- Notification reminders for video submissions and feedback availability.
- Motivational quotes to encourage users.
- Incentives for consistent practice and improvement.
- Option to connect with coaches (paid feature).
- Event creation for training sessions or competitions.
- Community support through user connections.

3.2. Functional Requirements

- Each user may only create one account.
- The app allows users to log in using their correct credentials.
- Users can customize their profiles.
- The app enables users to upload their bowling videos.
- Uploaded videos will be analyzed using AI/ML algorithms.
- The app identifies mistakes and generates feedback.
- Personalized training plans based on the analysis will be provided.
- Users can track and view their progress reports.
- Notifications will remind users about video submissions and feedback.
- Users can connect with coaches for additional guidance.
- The app facilitates event creation for training sessions.

3.3. Quality Attributes

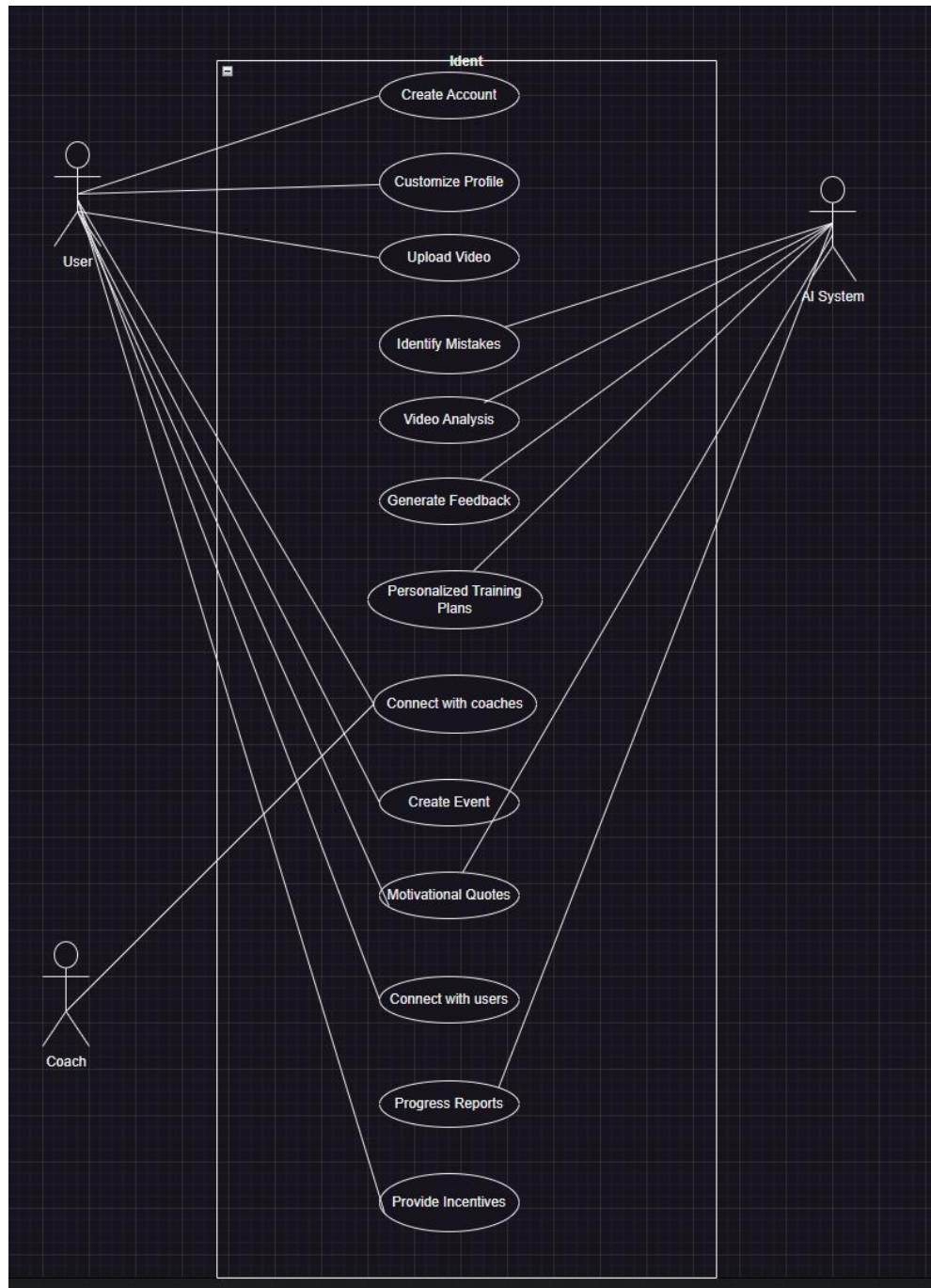
- **Affordability:** The application will be free to use, with optional paid features for coaching.
- **Debuggability:** Its modular structure makes it easy to identify and fix bugs.
- **Efficiency:** Optimized algorithms ensure quick video analysis.
- **Evolvability:** New features and improvements can be easily integrated based on user feedback.
- **Failure Transparency:** Users receive clear messages for any input errors during video upload.
- **Flexibility:** The system can adapt to changes in user requirements.
- **Maintainability:** The app will be built with tools that allow for easy updates.
- **Reliability:** Users will be notified of the success or failure of their actions.
- **Compatibility:** The app will be compatible with Android devices running version 6.0 or higher.
- **Responsiveness:** The app will respond effectively to user actions and device quality.
- **Reusability:** User interface components can be reused in future projects.

- **Security:** User data, including personal information and videos, will be securely stored.

3.4. Non-Functional Requirements

- Users should be able to log in within 5 seconds of entering the correct credentials.
- The system must verify credentials with the database before allowing login.
- Uploaded videos should be processed and analyzed promptly, providing feedback within 1-2 minutes.
- Users will receive notifications about new feedback within 10 seconds after analysis completion.
- The application must handle multiple users simultaneously without performance issues.
- Personal data must be encrypted and securely stored during transmission and storage.

4. Use Case Diagram



5. High Level Use case

5.1 Create Account

Use Case	Create Account
Actor	User/Player
Type	Primary
Description	User will create his account in our app.By adding necessary details including Name, email and phone number.

5.2 Customize Profile

Use Case	Customize profile
Actor	User
Type	Primary
Description	Users can customize their profile by adding more details including their age, experience , bowling style, injury if any etc.

5.3 Input Video

Use Case	Input video
Actor	User
Type	Primary
Description	User will input a video that has his bowling action so that it can further proceed to video analysis.

5.4 Identify mistakes

Use Case	Identify mistakes
Actor	User, AI coach
Type	Primary
Description	After the video is uploaded , the coach will analyze the video and identify the mistakes in the bowling technique or other mistakes the user

	has made.
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5.5 Generate Feedback

Use Case	Generate FeedBack
Actor	User, Ai coach
Type	Primary
Description	After identifying mistakes the Ai coach will give feedback i.e the solution and the correction based on the mistakes it had identified.

5.6 Create Personalized Plans

Use Case	Create Personalized plans
Actor	User
Type	Primary
Description	The user can make their plans and progress goals based on their ease and availability for example, next trial date and etc.

5.7 Track Progress Report

Use Case	Track Progress
Actor	User
Type	Primary
Description	Users can track their progress on the basis of what they performed before and on the way of improving their bowling journey.This helps them in improving more and stay motivated.

5.8 Get Notifications

Use Case	Get Notifications
Actor	User
Type	Primary
Description	The user can get notifications of any type for example, when they have

	a deadline to re-input video for analysis , or a friend sent a chat, etc.
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5.9 Get Motivational Quotes

Use Case	Get Motivational Quotes
Actor	User
Type	Primary
Description	Our app will provide users with motivational quotes so they will not lose pace and stay motivated throughout their journey. This will help them improve their game.

5.10 Connect with coach

Use Case	Connect to Coach
Actor	User
Type	Primary
Description	If the user wants to go one step further and needs some extra coaching then they can connect to specialized coaches. This feature will be provided by our app and it's paid.

5.11 Provide Incentives

Use Case	Provide incentives
Actor	Ai coach
Type	Primary
Description	The user will be incentivized by the admin based on their performance and time to time progress.

5.12 Create Events

Use Case	Create Events
Actor	User
Type	Primary

Description	Users can create events with other users using the app. For example they can set a match etc.
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5.13 Connect with other users

Use Case	Connect with users
Actor	User
Type	Primary
Description	Users can chat with other users, talk with them, share their feedback.

5.14 Manage Portal

Use Case	Manage Portal
Actor	Admin
Type	Primary
Description	Admin will manage the portal , he will add and remove users, add incentives, add coaches to the panel , etc.

6.Extended Use Case

6.1 Create Account

Use Case	Create Account
Scope	App-Khiladi
Level	User Goal
Primary Actor	User
Stakeholders	User: Who wants to create an account. Admin: Who will enable the user so that he can create an account.
Pre-Conditions	Users must have a stable internet connection. Admin must log in to validate the user.
Post-Conditions	Account created successfully and the dashboard is visible. Admin has an entry added with a new user.
Main Success Scenario	

	Actor Action	System Response
	1- User will open the app and navigate through it. 2- Click on Signup and enter credentials. 4- Account Created Successfully and dashboard is visible for the user.	3- System will record the user's data and create his account.
Extensions	The user didn't have a stable internet connection. Application lost its previous state, and the user has to re-enter all the data again. The user didn't enter the information right. Admin has not logged in the portal.	
Special Requirements	High speed internet connection. Simple and easy UI.	

6.2 Customize Profile

Use Case	Customize Profile			
Scope	App-Khiladi			
Level	User Goal			
Primary Actor	User			
Stakeholders	User: Who wants to customize his profile. Admin: Who will record the user data.			
Pre-Conditions	Users must have a stable internet connection. Admin must log in to validate the user. The user must be validated and verified by the admin.			
Post-Conditions	Profile has been customized successfully and the dashboard is visible. Admin has an updated version of user’s data.			
Main Success Scenario	<table><tr><td>Actor Action</td><td>System Response</td></tr></table>		Actor Action	System Response
Actor Action	System Response			

	1- User will open the app and navigate through it. 2- Click on the customized profile option and add details. 4- Profile customized Successfully and dashboard is visible for the user. 3- System will record all the user's data .
Extensions	The user didn't have a stable internet connection. Application lost its previous state, and the user has to re-enter all the data again. Admin has not logged in the portal.
Special Requirements	High speed internet connection. Simple and easy UI.

6.3 Input Video

Use Case	Input Video	
Scope	App-Khiladi	
Level	User Goal	
Primary Actor	User	
Stakeholders	User: Who wants to input the video. AI coach/System who will analyze that video.	
Pre-Conditions	Users must have a stable internet connection. Video is uploaded successfully.	
Post-Conditions	Video is uploaded successfully. System can detect the video for further processes.	
Main Success Scenario		
	Actor Action	System Response
	1- User will open the app and navigate through it. 2- Click on upload video and upload the video from camera or	

	gallery. 3- System will take the video and store it in a database. 4- Video uploaded successfully and loaded for further processing .
Extensions	The user didn't have a stable internet connection. Application lost its previous state, and the user has to re-input the data video. The video size is too big or too small. System in not accepting the video.
Special Requirements	High speed internet connection. Simple and easy UI.

6.4 Identify Mistakes

Use Case	Identify Mistakes
Scope	App-Khiladi
Level	User Goal
Primary Actor	AI model
Stakeholders	User: Who wants to identify his mistakes. AI Model: Which will be analyzing the video and identifying mistakes.
Pre-Conditions	Users must have a stable internet connection. Video has been uploaded successfully.
Post-Conditions	The AI model has identified all mistakes. User is able to see his generated mistakes.

Main Success Scenario		
	Actor Action	System Response/AI Model
	1- After the video is uploaded , the user is waiting for the model to identify mistakes. 3- User is able to see the mistakes identified by the model.	2- The Model has identified all the mistakes based on the video uploaded by user.
Extensions	The user didn't have a stable internet connection. Application lost its previous state, and the model has to reload the video again. The user didn't see any mistakes generated.	
Special Requirements	High speed internet connection. Simple and easy UI.	

6.5 Generate Feedback

Use Case	Generate FeedBack	
Scope	App-Khiladi	
Level	User Goal	
Primary Actor	AI Model	
Stakeholders	User: Who wants to generate FeedBack. AI Model: Generates the feedback based on the mistakes.	
Pre-Conditions	Users must have a stable internet connection. Mistakes are identified. User is able to see his mistakes.	
Post-Conditions	FeedBack generated successfully and the user is able to see the feedback.	
Main Success Scenario		
	Actor Action	System Response

	1- After the user is able to see his mistakes. 2- He will click on the Generate feedback option. 4-User is able to see the feedback generated by the model. 3- Ai model will generate the feedback based on the mistakes.
Extensions	The user didn't have a stable internet connection. Application lost its previous state, and the user is not able to see the feedback and now he will re-generate. Ai model will re-generate the feedback.
Special Requirements	High speed internet connection. Simple and easy UI.

6.6 Create Personalized Plans

Use Case	Create Account	
Scope	App-Khiladi	
Level	User Goal	
Primary Actor	User	
Stakeholders	User: Who wants to create an account. Admin: Who will enable the user so that he can create an account.	
Pre-Conditions	Users must have a stable internet connection. Admin must log in to validate the user.	
Post-Conditions	Account created successfully and the dashboard is visible. Admin has an entry added with a new user.	
Main Success Scenario		
	Actor Action	System Response
	1- User will open the app and navigate through it. 2- Click on Signup and enter credentials.	3- System will record the user's

	<table border="1"> <tr> <td>4- Account Created Successfully and dashboard is visible for the user.</td><td>data and create his account.</td></tr> </table>	4- Account Created Successfully and dashboard is visible for the user.	data and create his account.
4- Account Created Successfully and dashboard is visible for the user.	data and create his account.		
Extensions	The user didn't have a stable internet connection. Application lost its previous state, and the user has to re-enter all the data again. The user didn't enter the information right. Admin has not logged in the portal.		
Special Requirements	High speed internet connection. Simple and easy UI.		

6.7 Track Progress

Use Case	Track Progress	
Scope	App-Khiladi	
Level	User Goal	
Primary Actor	User	
Stakeholders	User wants to track his progress. Model will generate reports for his progress.	
Pre-Conditions	Users must have a stable internet connection. All the progress of the user is recorded in the database.	
Post-Conditions	Progress report is generated by model and user is able to track his progress.	
Main Success Scenario		
	Actor Action	System Response
	1- User will open the app and navigate through it. 2- From the menu, select the progress report. 4- User is able to track the	3- Report is generated by the system/model.

	<table border="1"> <tr> <td>progress over time.</td><td></td></tr> </table>	progress over time.	
progress over time.			
Extensions	The user didn't have a stable internet connection. Data is not recorded due to some technical issues and the report is not generated. Report is not being loaded.		
Special Requirements	High speed internet connection. Simple and easy UI.		

6.8 Get Notifications

Use Case	Get Notifications	
Scope	App-Khiladi	
Level	User Goal	
Primary Actor	User	
Stakeholders	User wants to get notifications. Admin will send notification to users about any event, from a friend`s chat etc.	
Pre-Conditions	Users must have a stable internet connection. Admin can send notifications.	
Post-Conditions	The user has received the notification.	
Main Success Scenario		
	Actor Action	System Response
	2-User has successfully received the notifications.	1- System has sent the notifications.
Extensions	The user didn`t have a stable internet connection. There is an issue in sending notifications due to a system technical issue. User is unable to receive notifications.	
Special Requirements	High speed internet connection.	

	Simple and easy UI.
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6.9 Get Motivational Quotes

Use Case	Get MotivationalQuotes					
Scope	App-Khiladi					
Level	User Goal					
Primary Actor	User					
Stakeholders	User wants to get motivational quotes. Admin will send motivational quotes to users.					
Pre-Conditions	Users must have a stable internet connection. User is logged in. Admin must log in.					
Post-Conditions	Motivational quotes are sent. Users are able to get them.					
Main Success Scenario	<table><tr><th>Actor Action</th><th>System Response</th></tr><tr><td>2- User is able to get motivational quotes.</td><td>1- System has sent motivational quotes.</td></tr></table>		Actor Action	System Response	2- User is able to get motivational quotes.	1- System has sent motivational quotes.
Actor Action	System Response					
2- User is able to get motivational quotes.	1- System has sent motivational quotes.					
Extensions	The user didn't have a stable internet connection. There is an issue in sending motivational quotes due to a system technical issue. User is unable to receive quotes.					
Special Requirements	High speed internet connection. Simple and easy UI.					

6.10 Connect with Coaches

Use Case	Connect with coaches.
Scope	App-Khiladi
Level	User Goal

Primary Actor	User				
Stakeholders	Users want to connect with specialized coaches. System will create setup and connect users with coaches. Coaches for training.				
Pre-Conditions	Users must have a stable internet connection. Admin must log in to validate the user. Coaches must be available for giving training.				
Post-Conditions	Users are connected with coaches and communication is ongoing.				
Main Success Scenario	<table border="1"> <thead> <tr> <th>Actor Action</th><th>System Response</th></tr> </thead> <tbody> <tr> <td> 1- User will open the app and navigate through it. 2- Select the connect with coaches option. 3-User will select the coach to which they want to connect. 5- Connection is successful and the user and coach are now communicating with each other. </td><td> 4- System will notify the specific coach and form a connection between user and coach. </td></tr> </tbody> </table>	Actor Action	System Response	1- User will open the app and navigate through it. 2- Select the connect with coaches option. 3-User will select the coach to which they want to connect. 5- Connection is successful and the user and coach are now communicating with each other.	4- System will notify the specific coach and form a connection between user and coach.
Actor Action	System Response				
1- User will open the app and navigate through it. 2- Select the connect with coaches option. 3-User will select the coach to which they want to connect. 5- Connection is successful and the user and coach are now communicating with each other.	4- System will notify the specific coach and form a connection between user and coach.				
Extensions	The user didn't have a stable internet connection. Coach is not available. Coach is unable to connect due to some technical issue. User is not able to connect. Disruption in communication.				
Special Requirements	High speed internet connection. Simple and easy UI.				

6.11 Provide Incentives

Use Case	Provide Incentives
Scope	App-Khiladi

Level	User Goal				
Primary Actor	Admin				
Stakeholders	Users will get incentives based on their performance. Admin will provide different incentives to users on the basis of their performance.				
Pre-Conditions	Users must have timely completed his tasks and performed well. Admin must log in to validate the user. Admin has all the progress reports of users.				
Post-Conditions	Incentives are provided successfully by the admin and received by the users.				
Main Success Scenario	<table border="1"> <thead> <tr> <th>Actor Action</th><th>System Response</th></tr> </thead> <tbody> <tr> <td>2- User has received incentives.</td><td>1- System has provide incentives based on the user's performance.</td></tr> </tbody> </table>	Actor Action	System Response	2- User has received incentives.	1- System has provide incentives based on the user's performance.
Actor Action	System Response				
2- User has received incentives.	1- System has provide incentives based on the user's performance.				
Extensions	The user didn't have a stable internet connection. Admin has not logged in the portal. Admin does not have a record or a particular user. User did not perform well.				
Special Requirements	High speed internet connection. Simple and easy UI.				

6.12 Create Events

Use Case	Create Events
Scope	App-Khiladi
Level	User Goal
Primary Actor	User
Stakeholders	User wants to create an event with other users. Admin is responsible for setting up the event.
Pre-Conditions	Users must have a stable internet connection. Users must log in to their account. Admin must log in to validate the user.

Post-Conditions	Event is created successfully, Users are communicating with each other.	
Main Success Scenario		
	Actor Action	System Response
	1- User will open the app and navigate through it. 2- Creates an event by sending requests to others. 5- Event dashboard is shown to all users.	3- System will notify other users about the event. 4- Create an event.
Extensions	The user didn't have a stable internet connection. Application lost its previous state, and the user has to re-create an event. System has some technical issues. Users are not available.	
Special Requirements	High speed internet connection. Simple and easy UI.	

6.13 Connect with users

Use Case	Connect with users	
Scope	App-Khiladi	
Level	User Goal	
Primary Actor	User	
Stakeholders	User wants to connect with other users. Admin will allow this communication by notifying other users.	
Pre-Conditions	Users must have a stable internet connection. Admin must log in to validate the user. All users get notified.	
Post-Conditions	Event created successfully and the dashboard is visible to all users.	
Main Success Scenario		
	Actor Action	System Response

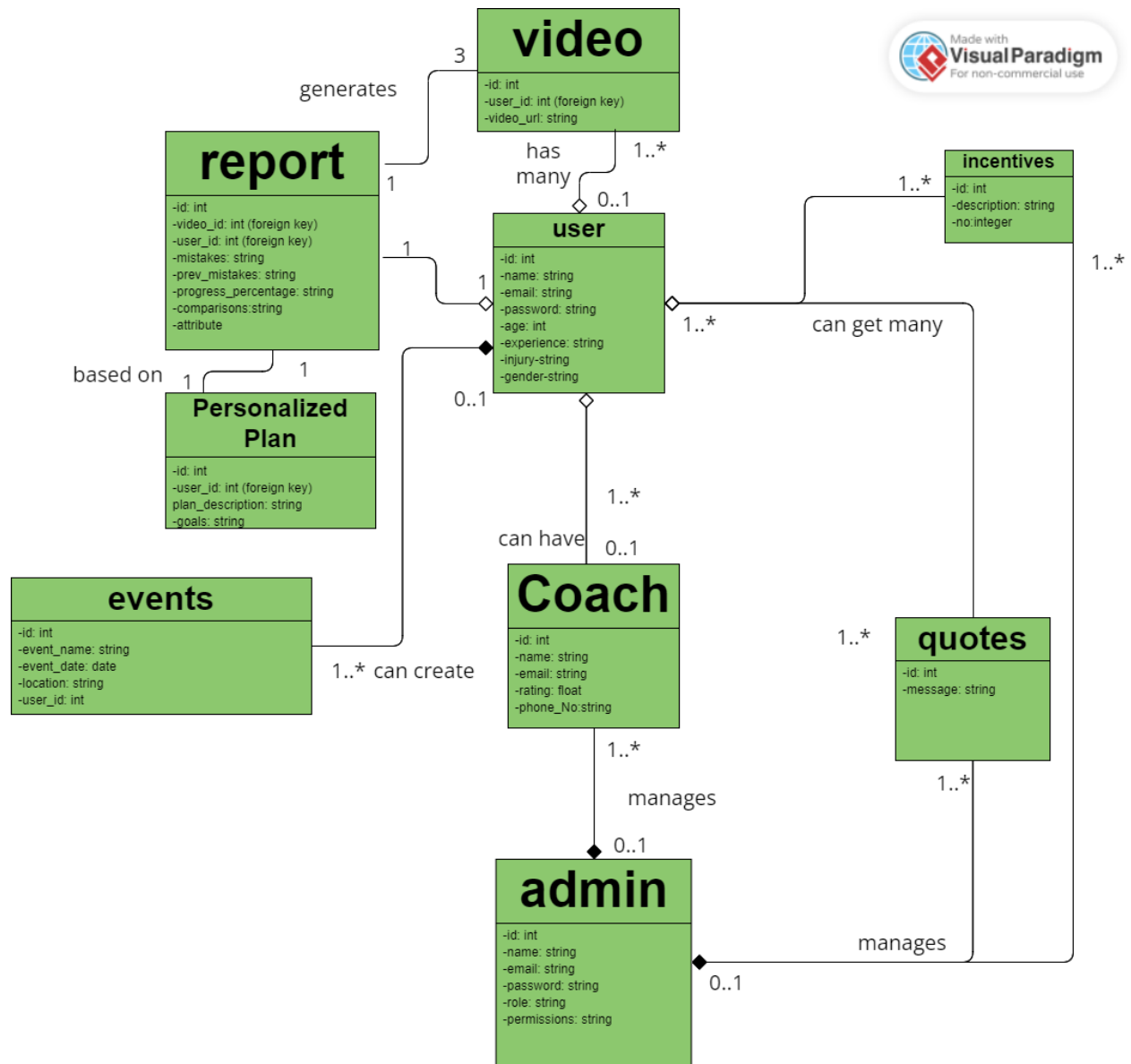
	1- User will open the app and navigate through it. 2- User creates an event and adds other users. 4-Event created successfully and the same dashboard is shown to all users.. 3-System will notify other users and make an event.
Extensions	The user didn't have a stable internet connection. Event not created due to some technical issues. Other users are not available. Admin has not logged in the portal.
Special Requirements	High speed internet connection. Simple and easy UI.

6.14 Manage Portal

Use Case	Manage Portal		
Scope	App-Khiladi		
Level	Admin Goal		
Primary Actor	Admin		
Stakeholders	Admin wants to manage the app, add or remove users and incentives, etc.		
Pre-Conditions	Admin must have a stable internet connection. Admin must log in to the app. Admin has all credentials.		
Post-Conditions	Admin is able to make any changes to the app or update the app..		
Main Success Scenario	<table> <tr> <td>Actor Action</td><td>System Response</td></tr> </table>	Actor Action	System Response
Actor Action	System Response		

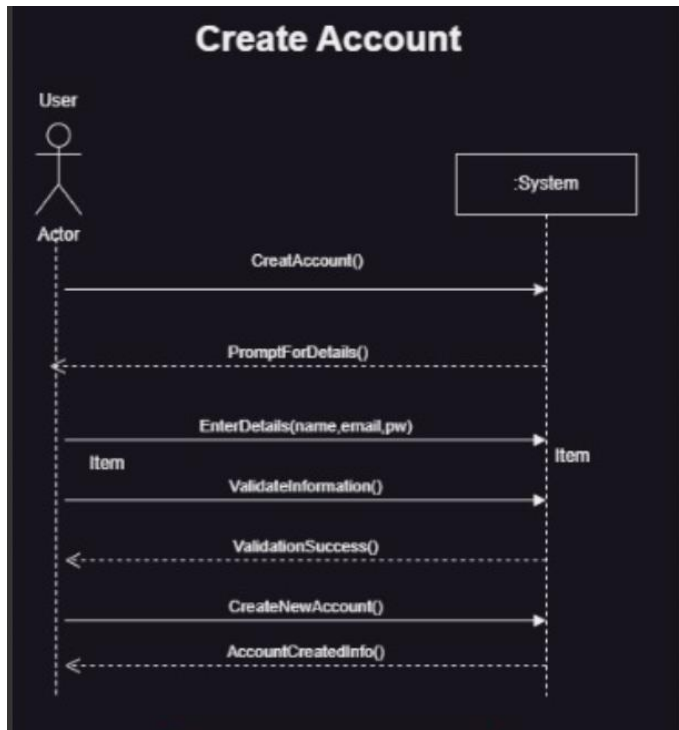
	1- Admin will login to the account. 2- Admin can make any changes or update the app. 4- Updates visible to admin. 3- System will record and update any changes if made by the admin.
Extensions	The Admin didn't have a stable internet connection. Application lost its previous state, and the admin has to make changes again. Admin has not logged in the portal.
Special Requirements	High speed internet connection. Simple and easy UI.

7. Domain Model

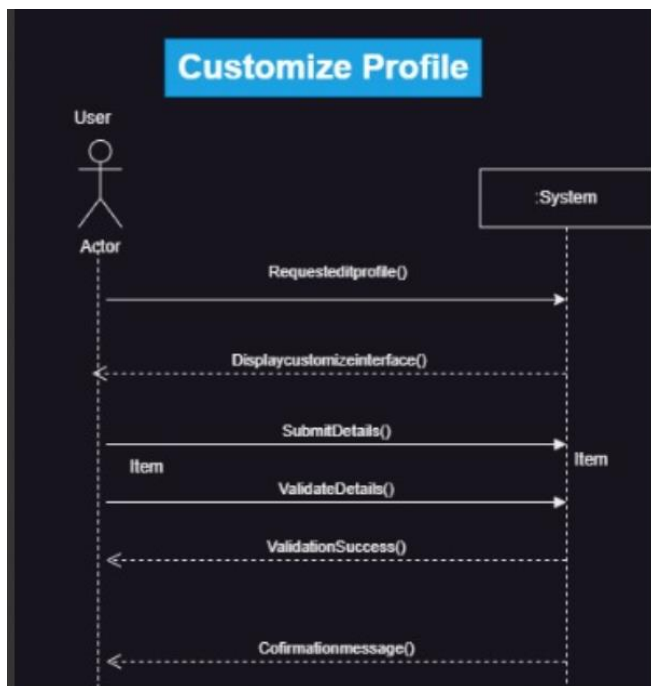


8. System Sequence Diagram

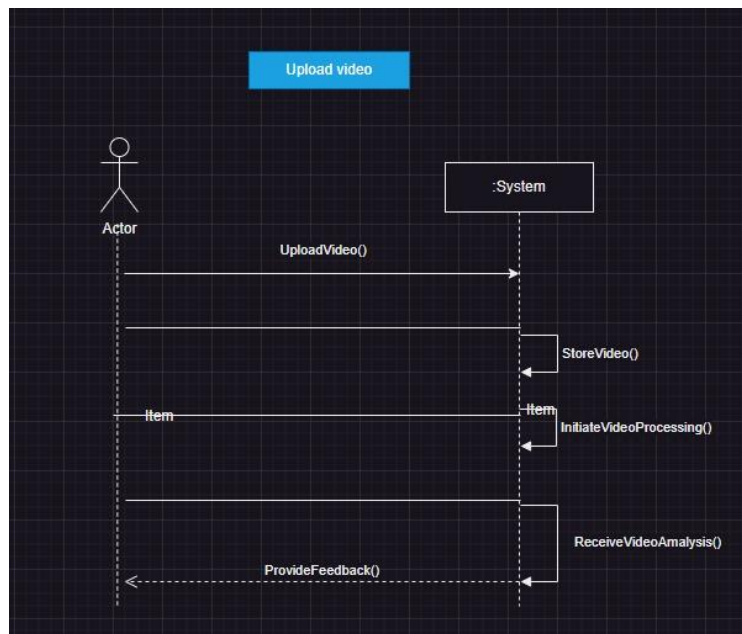
8.1 Create Account



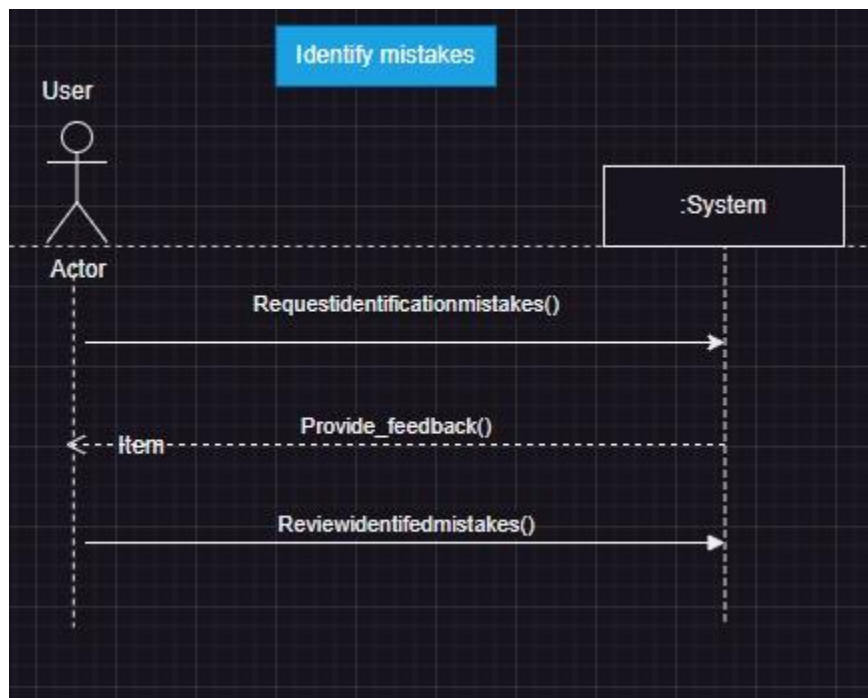
8.2 Customize Profile



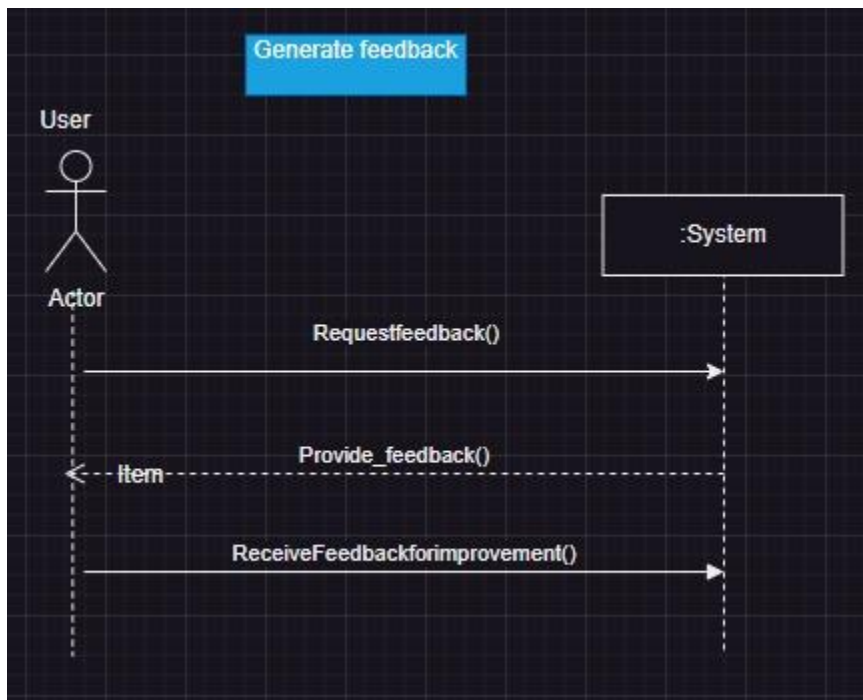
8.3 Upload Video



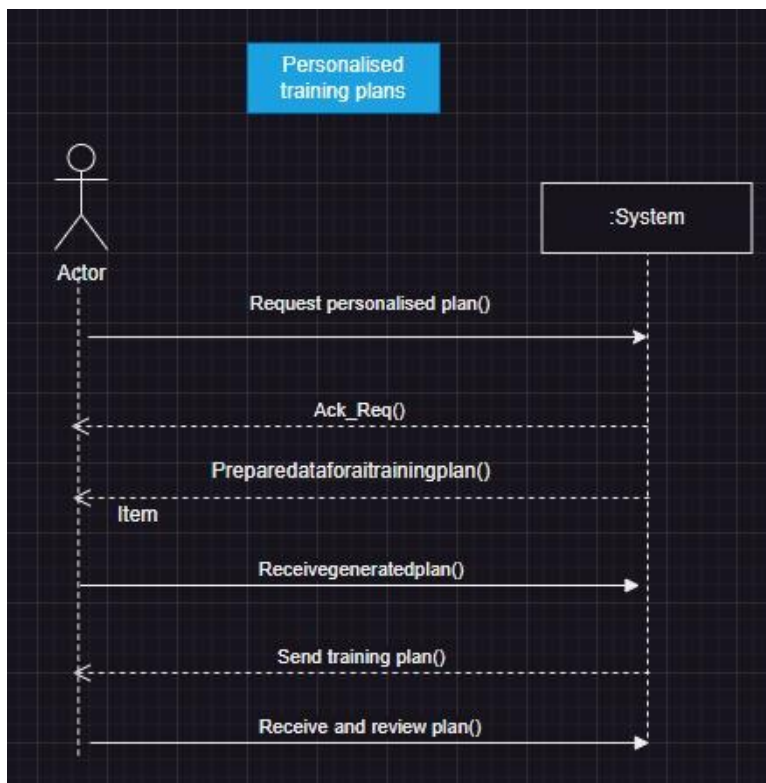
8.4 Identify Mistakes



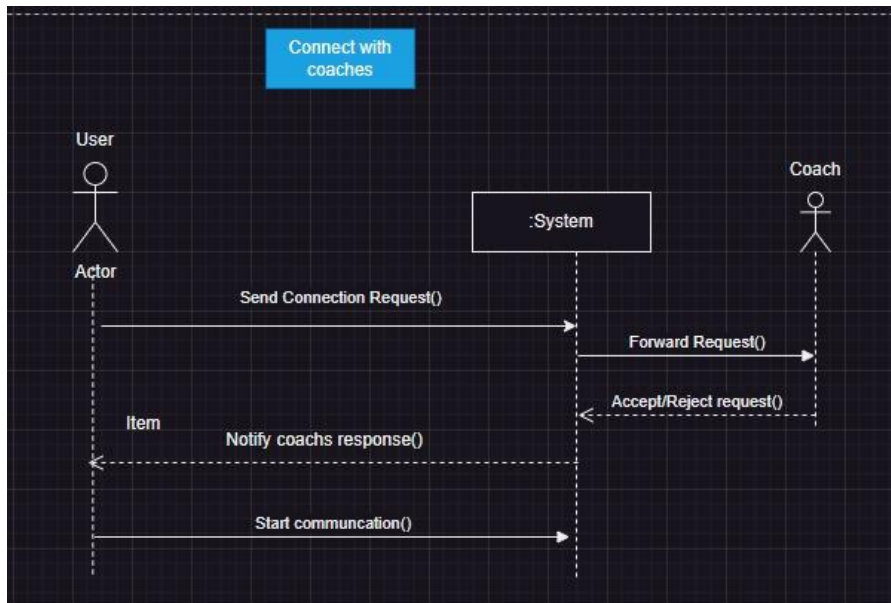
8.5 Generate Feedback



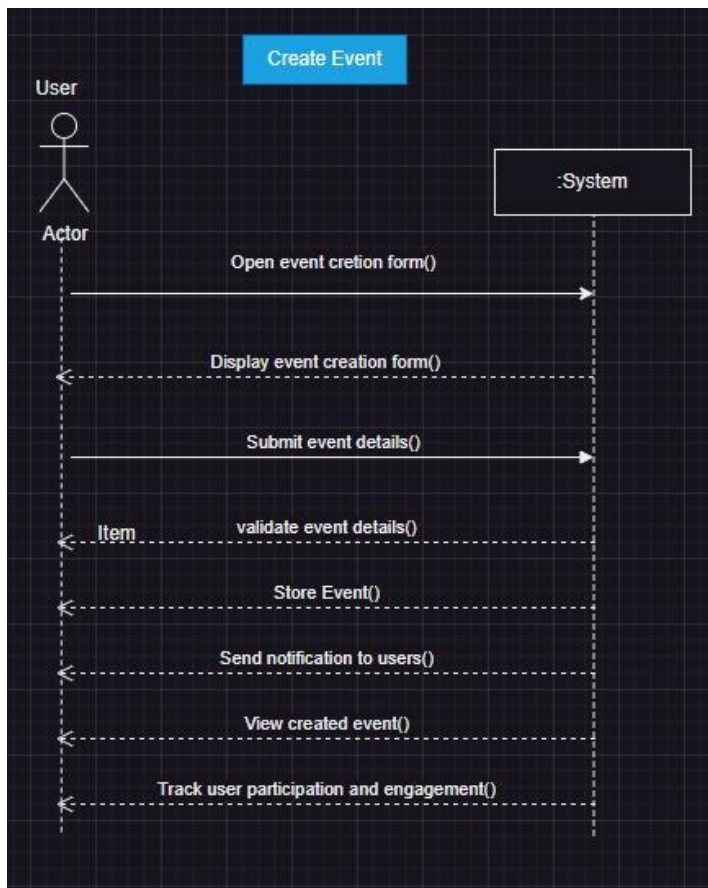
8.6 Personalized Training Plans



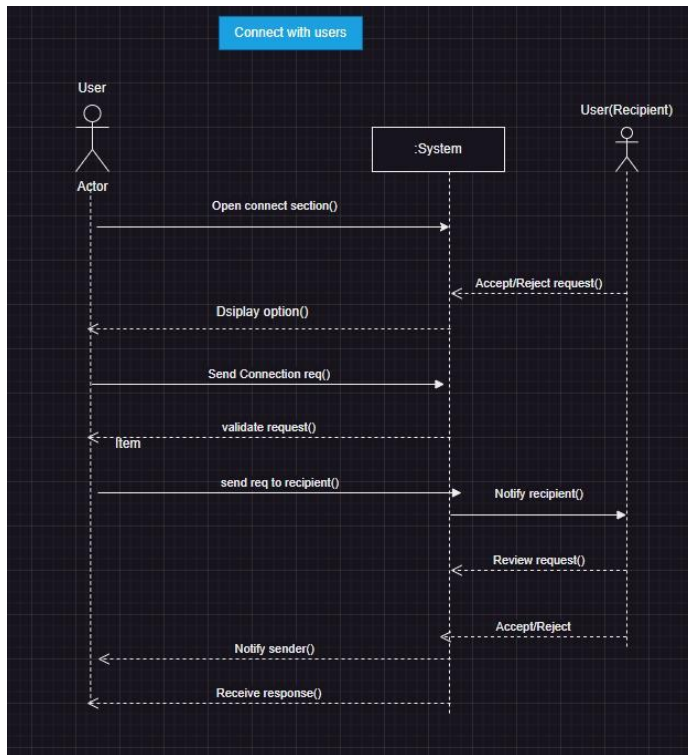
8.7 Connect With Coaches



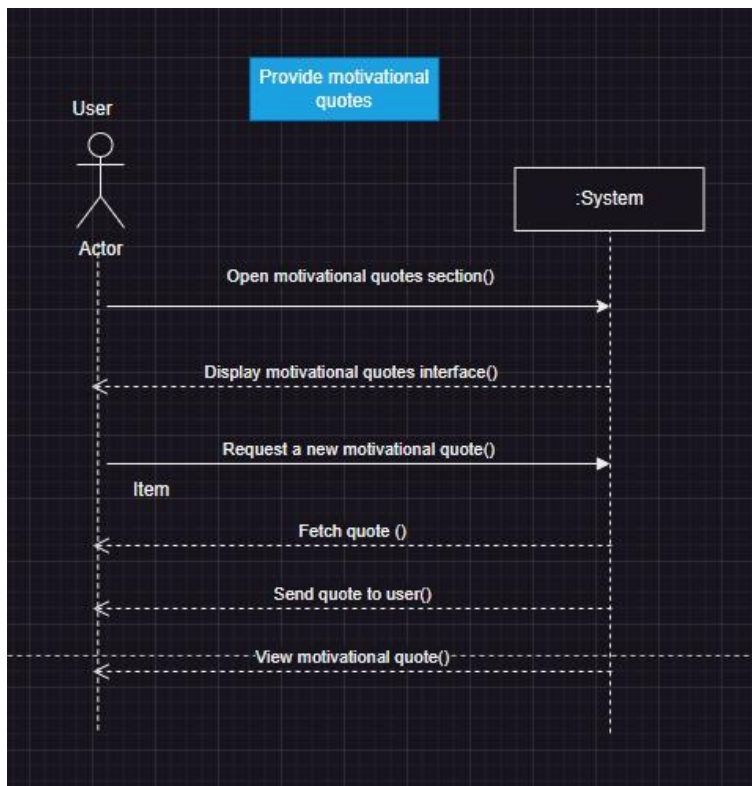
8.8 Create Events



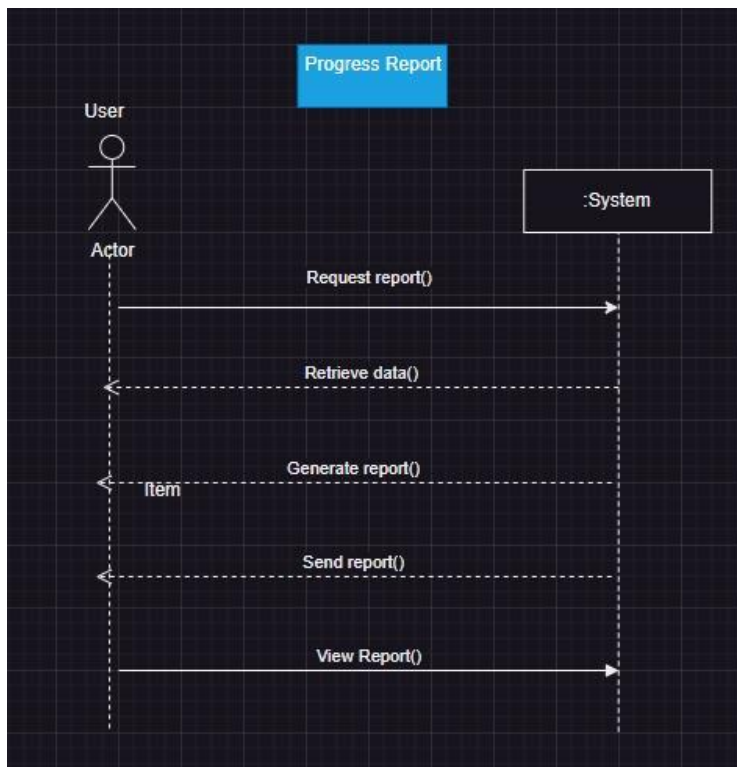
8.9 Connect with users



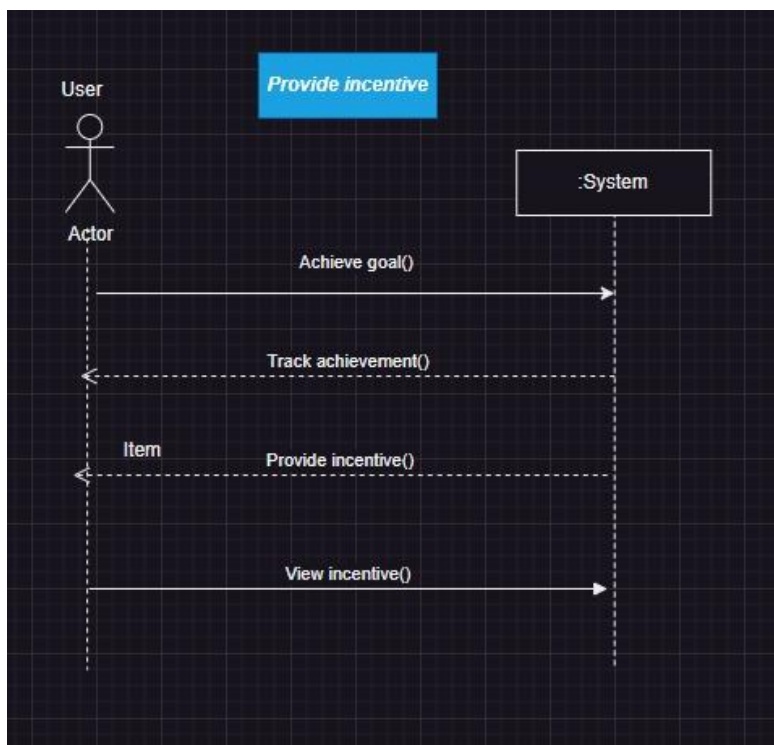
8.10 Provide Motivational Quotes



8.11 Generate Progress Report

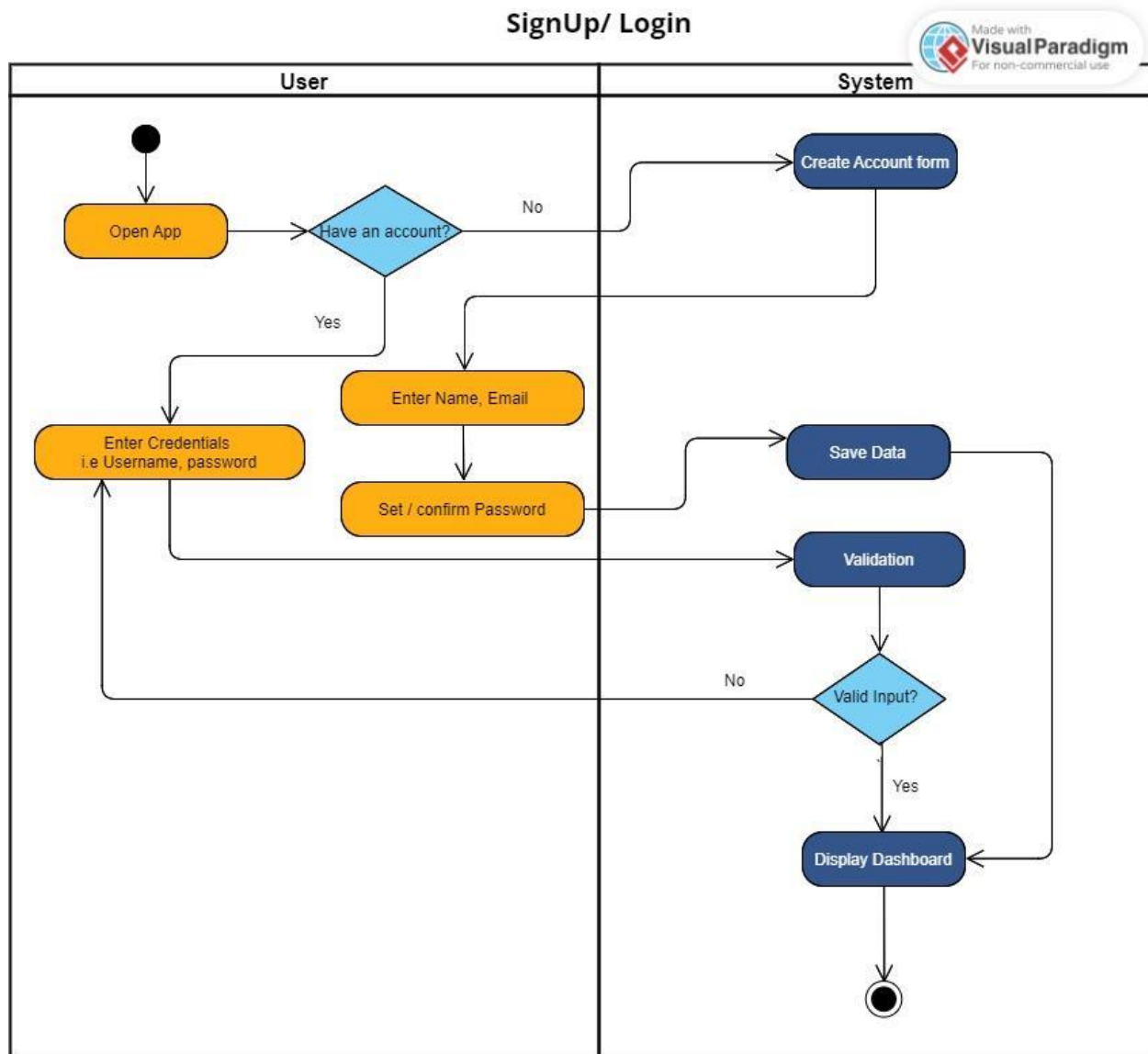


8.12 Provide Incentives

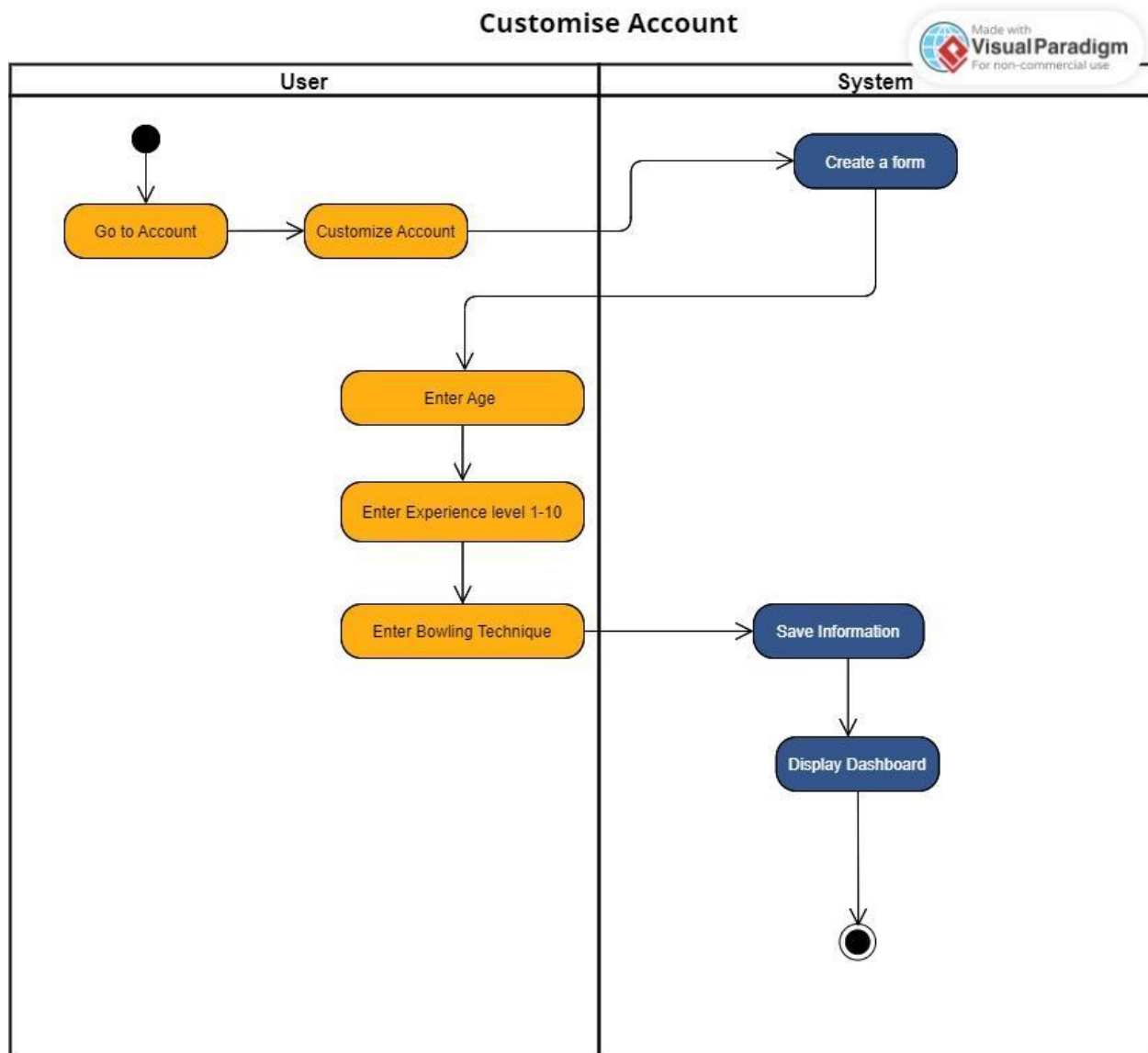


9. Activity Diagram

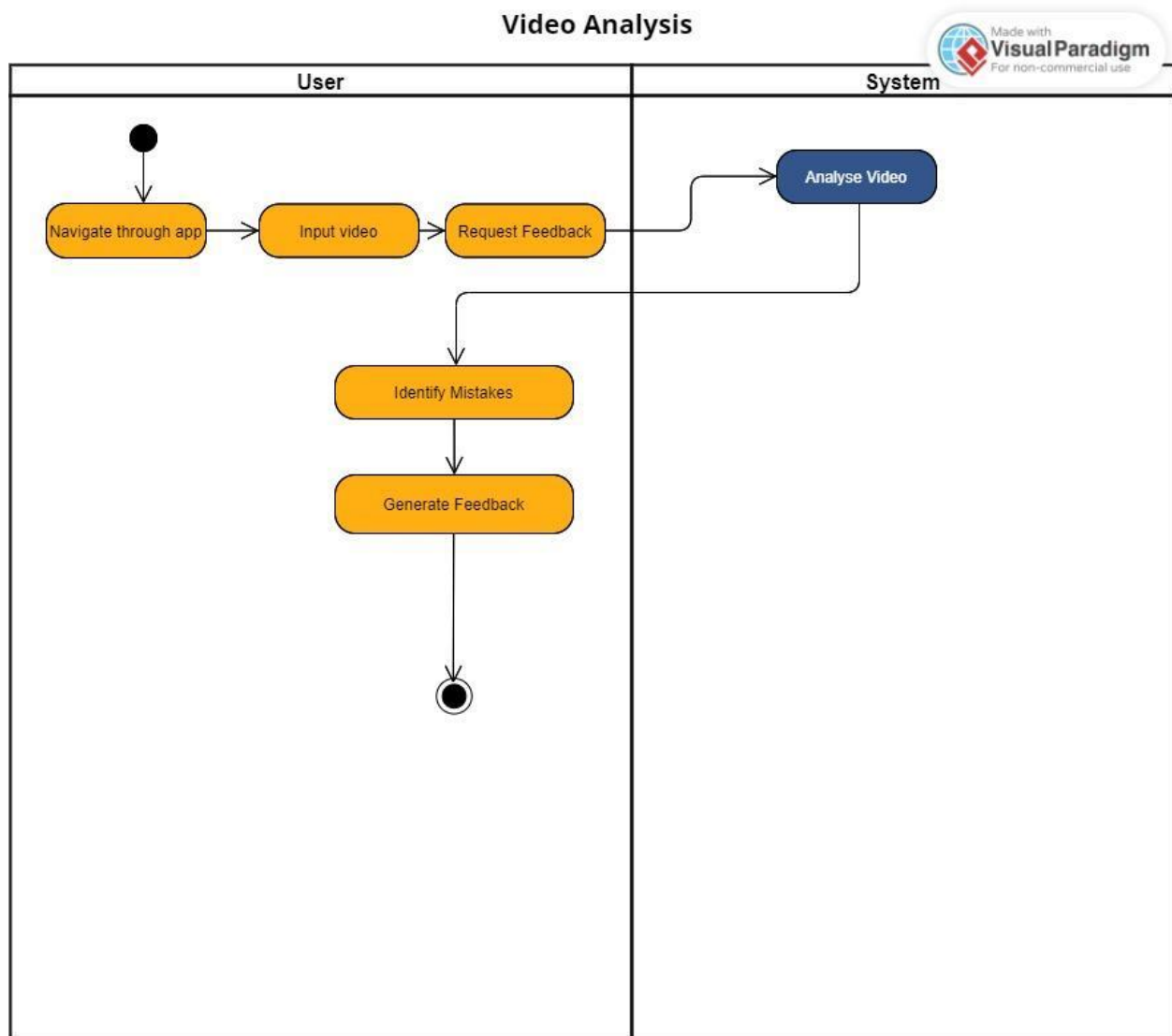
9.1 Create Account



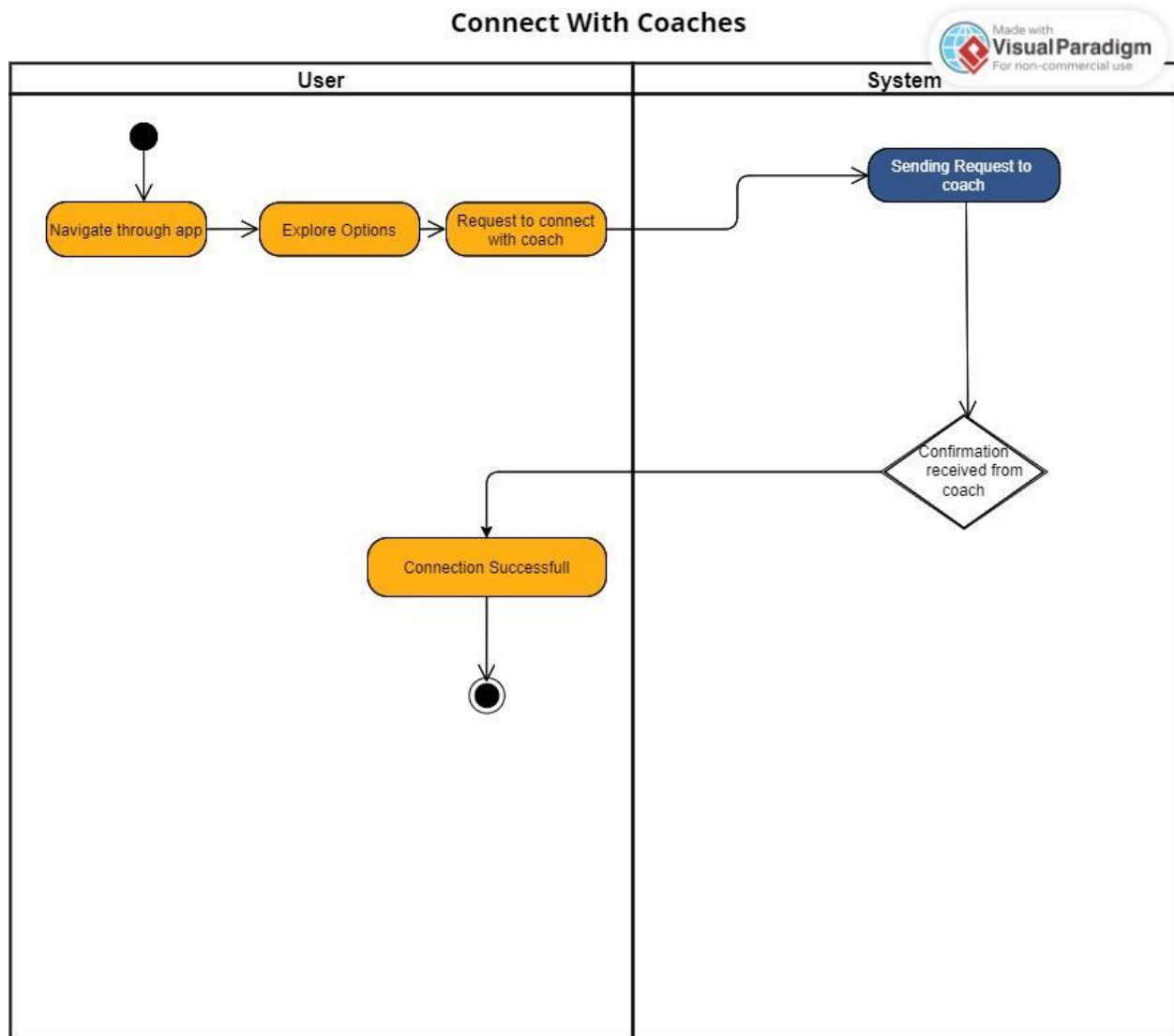
9.2 Customize Account



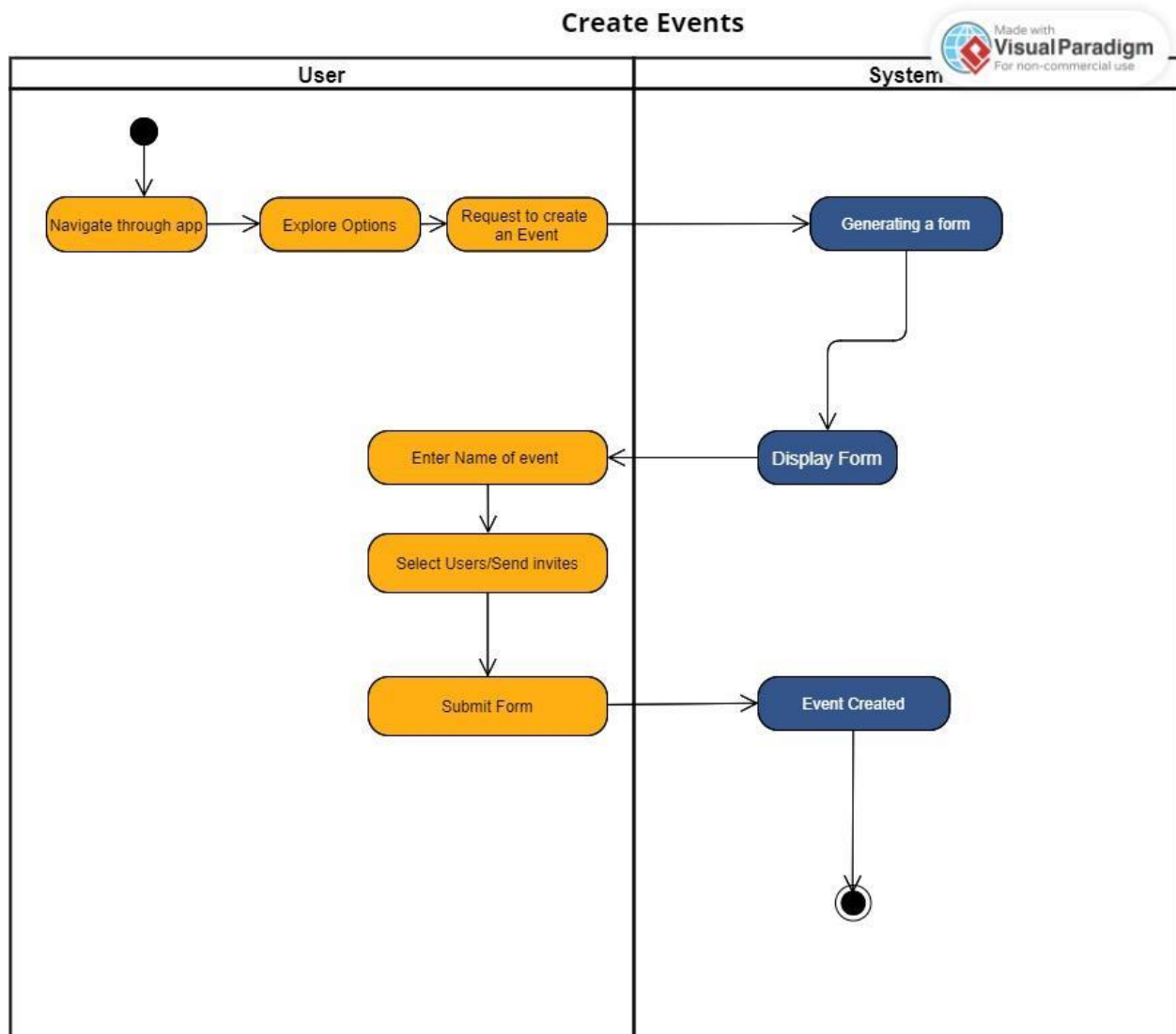
9.3 Video Analysis



9.4 Connect with coaches



9.5 Create Event



10. Operations Contracts

10.1 Create Account:-

Operation: CreateAccount()

Cross-references: Use Case – Create Account

Pre-conditions:

1. User needs an internet connection always on while using the application.
2. Users must not be already logged into an account, otherwise this should be the home page only.
3. Users are expected to give personal details as regards to the account (e.g., name, email address, password).
4. It should also be active and should be able to take new account creation requests.

Post-conditions:

1. Default the credit status in the new account created for the user in the system.
2. The name, e-mail and encrypted password are stored into the database to ensure the user's data's confidentiality.
3. A new account is created, and a unique user ID is assigned to differentiate the new user from the other users.
4. The user is notified or emailed about creation of the account.
5. The event of account creation is documented in the system thereby enabling future reference.

10.2 Customize Profile:-

Operation: CustomizeProfile()

Cross-references: Use Case – Customize Profile

Pre-conditions:

1. User requires an internet connection for the application to work properly at all times.
2. A user should be logged into their account before using the MOB/Panel.
3. Here an important assumption is more stipulated – the user profile in the social network must already be created.

Post-conditions:

1. Added new details and the profile of the user is current.
2. All the updated profile details are saved, and respectively, are retained in the database.

3. System informs user that his/her profile update is complete/confirmed.

10.3 Input Video

Operation: InputVideo()

Cross-references: Use Case – Upload Video

Pre-conditions:

1. Internet connection is required by the user.
2. On the web, it is required that the user must be logged into his account.
3. The some video file is in a supported format.

Post-conditions:

1. This video is uploaded in the system and remain as part of the system.
2. A record of the video is stored in the database.
3. The system congratulates the user for a successful upload of the video.

10.4 Identify Mistakes

Operation: IdentifyMistakes()

Cross-references: Use Case – Identify Mistakes

Pre-conditions:

1. User must have uploaded a Video.
2. The video must be uploaded and saved in order to be used in the system.
3. For AI Video Analysis it is mandatory to be processed.

Post-conditions:

1. The system of computer vision helps to evaluate the video for the possible mistakes.
2. It is possible to track down incorrect approaches which include but are not limited to; posture and technique errors.
3. Feedback is created and then delivered to the user in the form of feedback.

In any case, a record is kept for further use.

10.5 Generate Feedback

Operation: GenerateFeedback()

Cross-references: Use Case – Generate Feedback

Pre-conditions:

1. User must have performance data, for instance in the form of video analysis and feedback results.
2. User must have an account created and logged in before they are enabled to use the application.
3. It is crucial that The AI Feedback Engine should be on.

Post-conditions:

1. The performance data is transferred to the AI Feedback Engine of the system.
2. Culturally-mediated feedback is obtained from the performance data needed for the generation of personalized feedback.
3. Feedback is sent to the user.
4. The feedback received can be retrieved by the system at a later date if it is required.

10.6 Create Personalized Plans

Operation: CreatePersonalizedPlan()

Cross-references: Personalized Training Plans Use Case

Pre-conditions:

1. User must have an account and must sign in to it.
2. The user profile should contain the necessary data (for example, the skill level, the goals).
3. It means that the AI Training Plan Generator needs to be running.

Post-conditions:

1. User data is transferred to the AI Training Plan Generator.
2. The training plan for an individual employee is developed.
3. From here, the training plan is presented to the user.
4. Also, the plan is saved in the system in case of further modification or change of content.

10.7 Track Progress Report

Operation: TrackProgressReport()

Cross-references: Here is one of the Use Cases that the software developed, through the Object-Oriented Analysis and Design to serve the Organization:

Pre-conditions:

1. User cannot be a guest, they must be logged into his or her account.
2. The system must also possess information regarding the performance of the same in the past when it is accessible to the user.
3. The progress tracking module is a necessity and should be an up-and-running one.

Post-conditions:

1. The system pulls out data of performance of the user and uses this data to come up with another set of data.
2. It is from this analysis that a progress report is produced.
3. The report is given to the user.
4. It is kept for future use so that one can get extra information and compare the new results with the previous ones that were obtained from the report.

10.8 Get Notifications

Operation: GetNotifications()

Cross-references:

To this end, this operation contributes to the notification management use case designed to provide the user with information about events and changes in the system, as required.

Pre-conditions:

1. To qualify for notifications, the user must be logged into their account as delivery of notifications is personalized.
2. The system should have individually relevant notifications i.e. new messages, updates, events etc should be available to the user.
3. The Notifications module should always be active and effectively working all the time.

Post-conditions:

1. User notifications he/she requires are obtained by the system from the database.
2. The notifications are provided to the user at the right time with the help of the interface provided with more capabilities of delivering the notifications with the help of media like e-mail, push notifications.
3. They are labeled as read and unread depending on the actions of the user of the site.

4. Notification status information is saved for future use, it also allows the user to look at the notification's appearance later.

10.9 Get Motivational Quotes

Operation: GetMotivationalQuotes()

Cross-references:

This operation aids the feature that gives quotes with an intention of inspiring users time and over again whenever they are employing the system.

Pre-conditions:

1. It would require the users to be logged into their account to view such, motivational or relevant quotes to them.
2. The system should have a list of motivational quotes on record so that it can make it available for use.
3. The motivational quotes module has to be live and running as it opens and works at the start of each day.

Post-conditions:

1. The system will access motivational quotes in the database or a repository.
2. The quote selected finally is presented to the user of the system through the terminal or through the channel designated to the notification services.
3. The system follows a list of quotes it presents to the user to prevent redundancy and give a wide variety.
4. The quote is saved to the session or history of the user, so the user can always return to it.

10.10 Connect with Coach

Operation: ConnectWithCoach()

Cross-references:

This operation facilitates the 'with coach use case' where a user is in touch with a coach probably for some advice or training.

Pre-conditions:

1. The user must be on the application and signed into his/her account, the user must also have permissions to connect with a coach or better yet must be on the premium subscription level.
2. The system has to provide available coaches that are willing to accept connection requests.
3. The communication channel that links users and coaches has to be open.

Post-conditions:

1. The system makes a request to the selected coach for the user.
2. With the connection request, the notification gets to the coach who can either accept or deny the connection.
3. If accepted, the system creates communication between the user and the coach (for example, in the chat or video session).
4. The connection status together with the interaction history are recorded for use in the future.
5. If the request is declined then the system informs the user and gives other options like giving other available coaches or trainers.

10.11 Provide Incentives

Operation: ProvideIncentives()

Cross-references:

This operation enforces the use case used to provide incentives on top of achieved goals or activity within the system.

Pre-conditions:

1. It will be only available for the user enrolled in an account, and the user will need to meet certain achievement requirements during the usage of the app (e.g., the user has performed x amount of tasks, achieved y milestones).
2. The reward articles must also undergo a logical, relevant, and present accomplishments and possible rewards database.
3. The rewards calculation module needs to be running and set up properly to check for the eligibility and then calculate for the right incentives.

Post-conditions:

1. It computes the type and quantity of incentive converging with the rules set by the system through a user's last activities and performance.
2. It also records every stage of the process of the calculation for audit and troubleshooting. This comprises user's ID, criteria that is achieved, incentive arriving at; and date/time the operation is completed.

3. It is credited to the user's account, further, the user receives a notification that the incentive has been added to his/her account through the selected communication method such as, system notification and email.
4. The system modifies the user's rewards history in its database by adding the new incentive for the user to track in the future.
5. On this, the system captures the completion of the incentive provision and any possible errors to the process.

10.12 Create Events

Operation: CreateEvent()

Cross-references:

This operation enables a user for instance admin or event creator to initiate and invite other users, events that can be set at a certain time.

Pre-conditions:

1. The user must have an account and have authorisation to create events for the account.
2. The system must have a contemporary data base where all event details should be stored.
3. The event creation module has to be working, these are time slots and event categories have to be available.

Post-conditions:

1. Information such as the event name, description, date, time and participants is then recorded and stored in the event database of the system.
2. The system checks some required input data against the corresponding event data to check for the correctness of the input data and against the real events to avoid scheduling conflict.
3. All the related users of the event creation are informed through notification or email alerts.
4. The system also records the event creation for audit purpose the event creator details, time and, error if any occurred.
5. After being posted, the event is accessible to other users where they can observe, contribute to, or subscribe to events according to their roles or subscriptions.
6. It keeps record of user activity and interaction level to the event.

10.13 Connect with other users

Operation: ConnectWithUsers()

Cross-references:

This operation allows the users to be able to communicate to and from other users in the system enhancing the interaction.

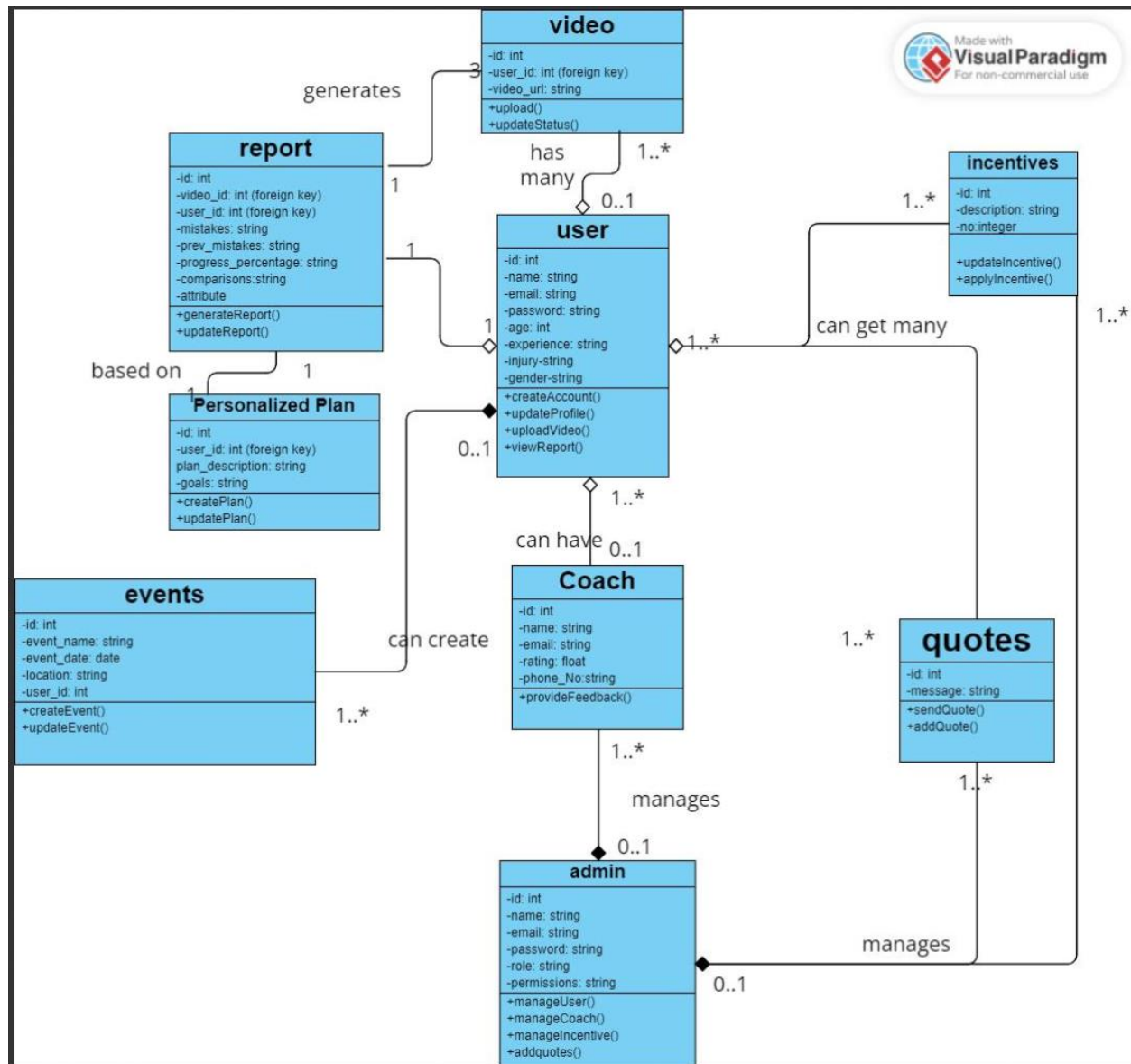
Pre-conditions:

1. In the current interface, the user must be logged into his/her account.
2. The system will have to keep contact permissions of the users and ensure that privacies are well observed.
3. The messaging/communication module must be initiated and set up to perform user to user communications.

Post-conditions:

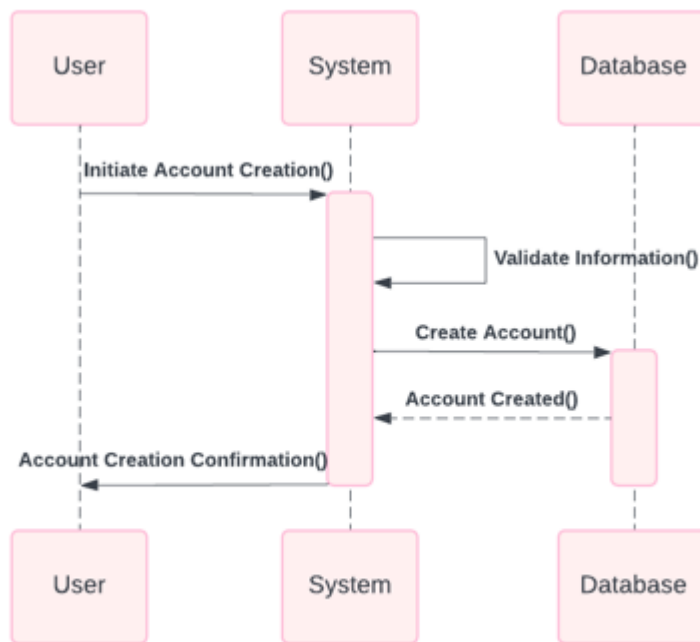
1. The system then checks the connection request to ensure that both users want the connection as per their privacy or accounts preferences.
2. Recipient receives the connection request and can either accept it or deny the connection request.
3. On successful connection the system records the adjusted relationship status of both users in the system upon acceptance of the request.
4. The system contains a communication platform where people can send messages or engage one other.
5. Whenever a connection request is refused the system records the action and at the same time alerts the concerned user.
6. Requests, acceptance and rejection of connection are recorded in the system for audit and monitoring.
7. Both users can now interact and communicate with each other through messages embedded in the system or through built-in collaboration tools.

11. Class Diagram

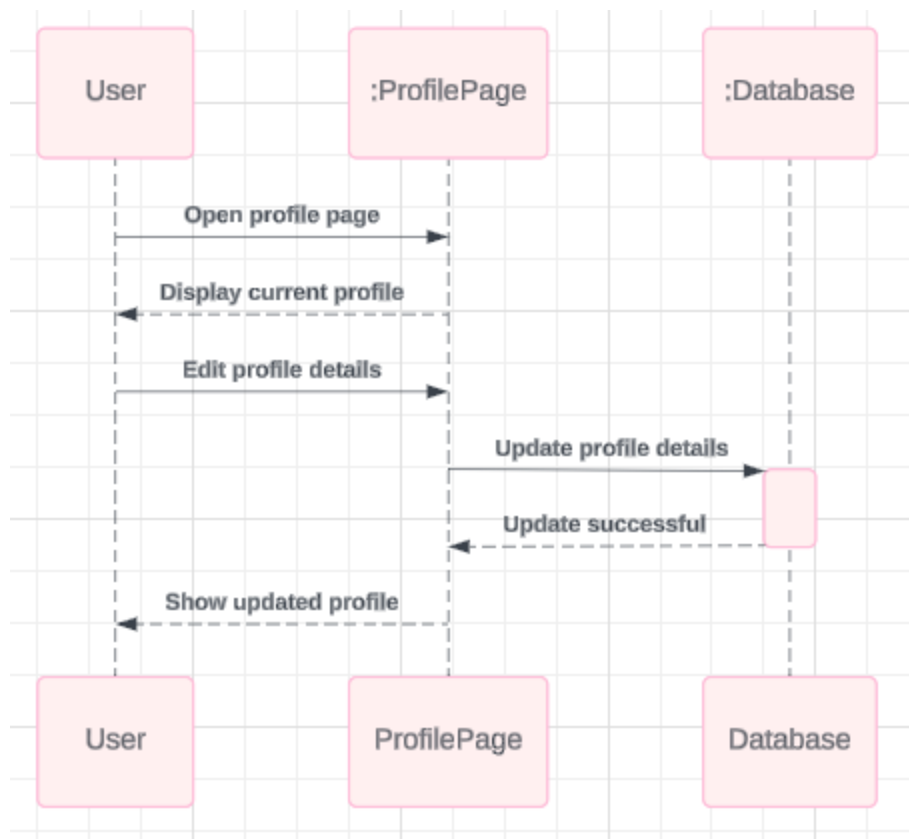


12. Sequence Diagram:-

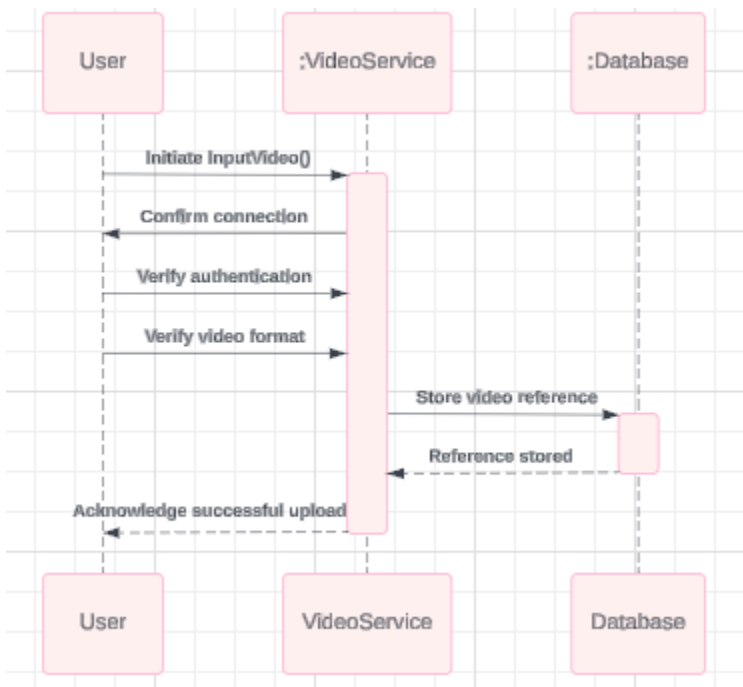
12.1 Create Account:



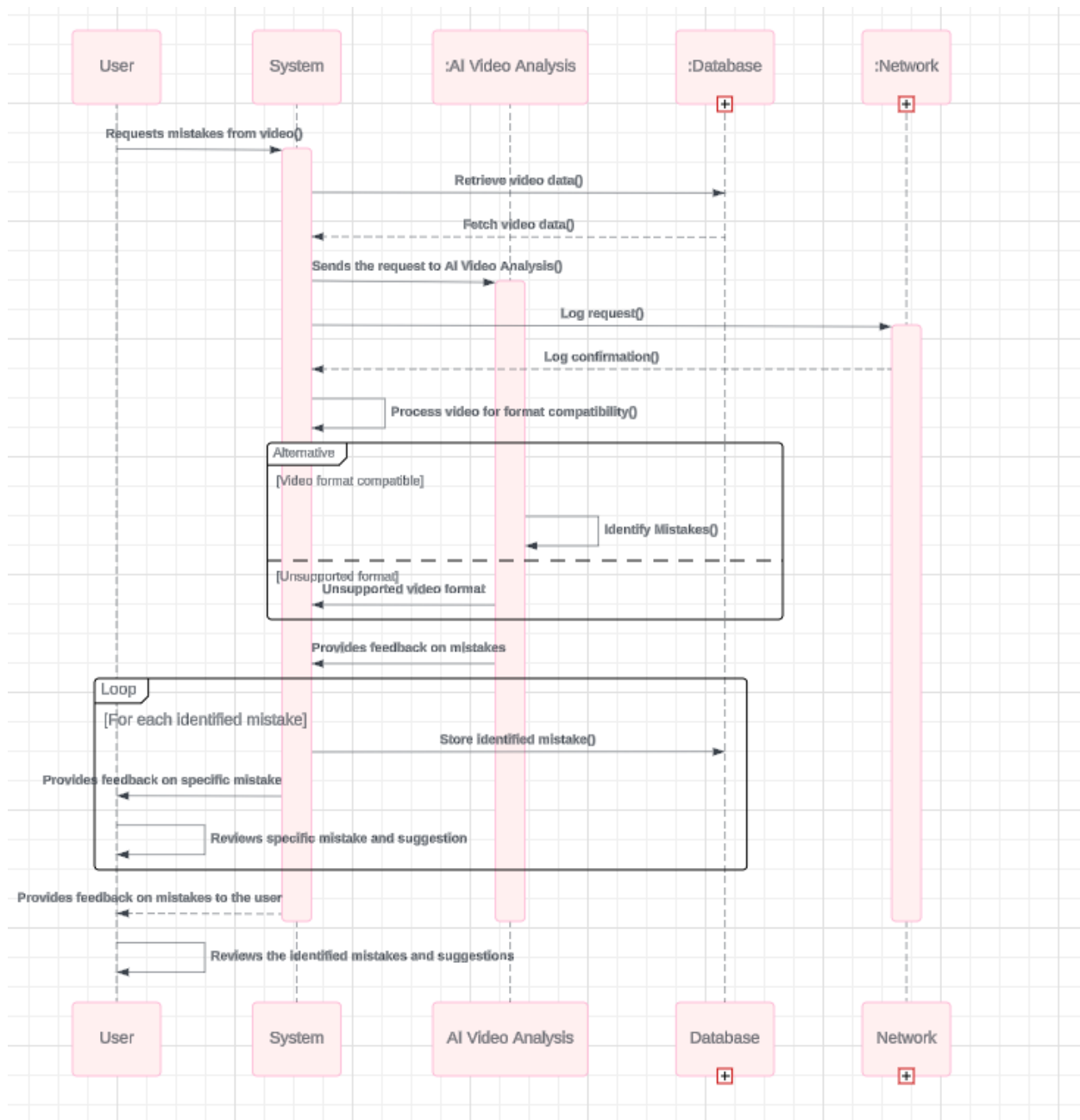
12.2 Customize Profile



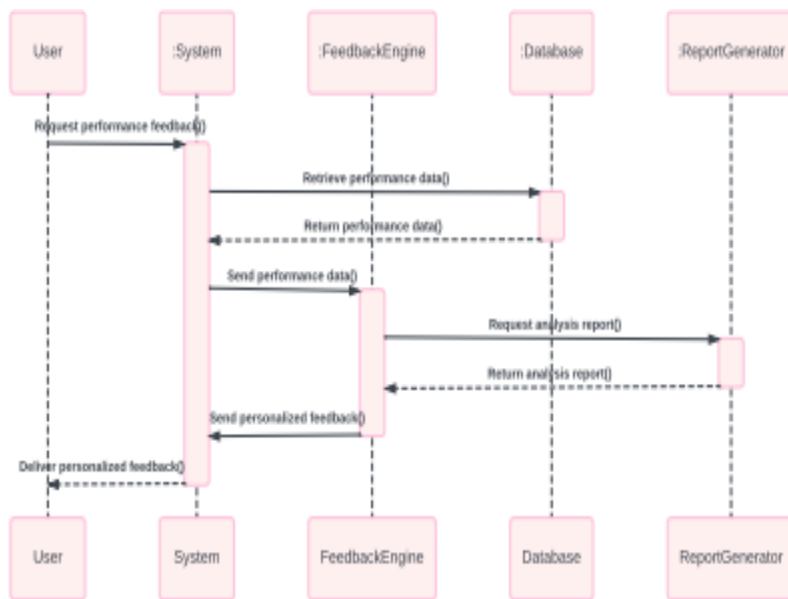
12.3 Input Video



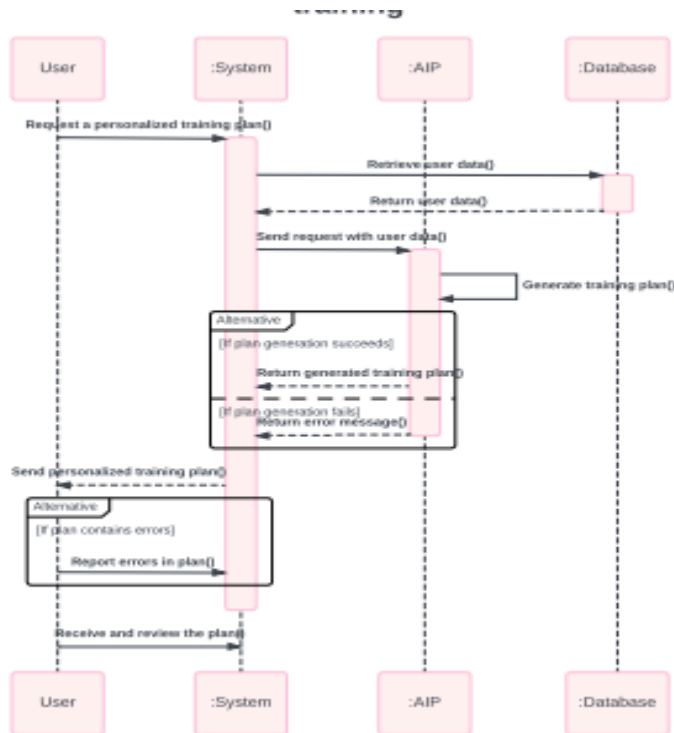
12.4 Identify Mistakes



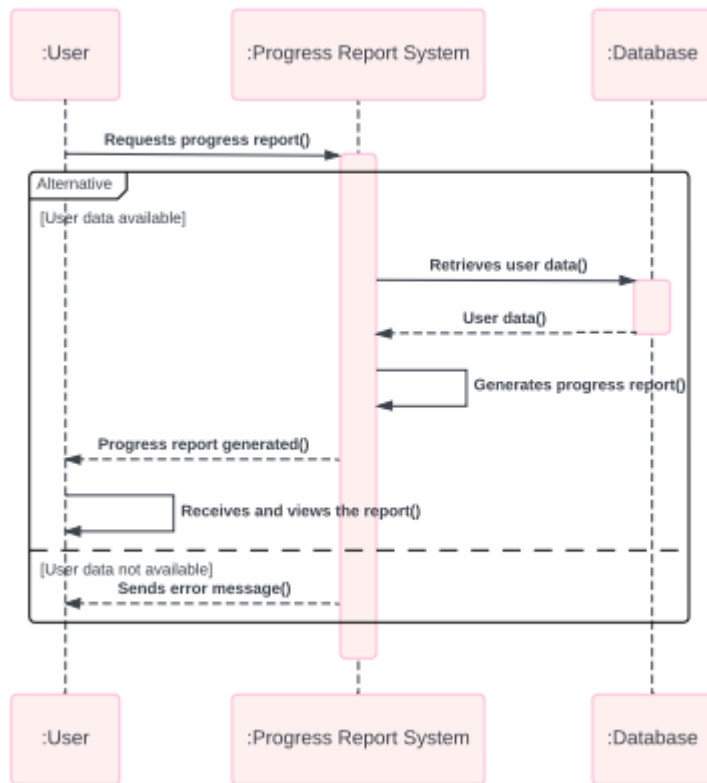
12.5:Generate Feedback



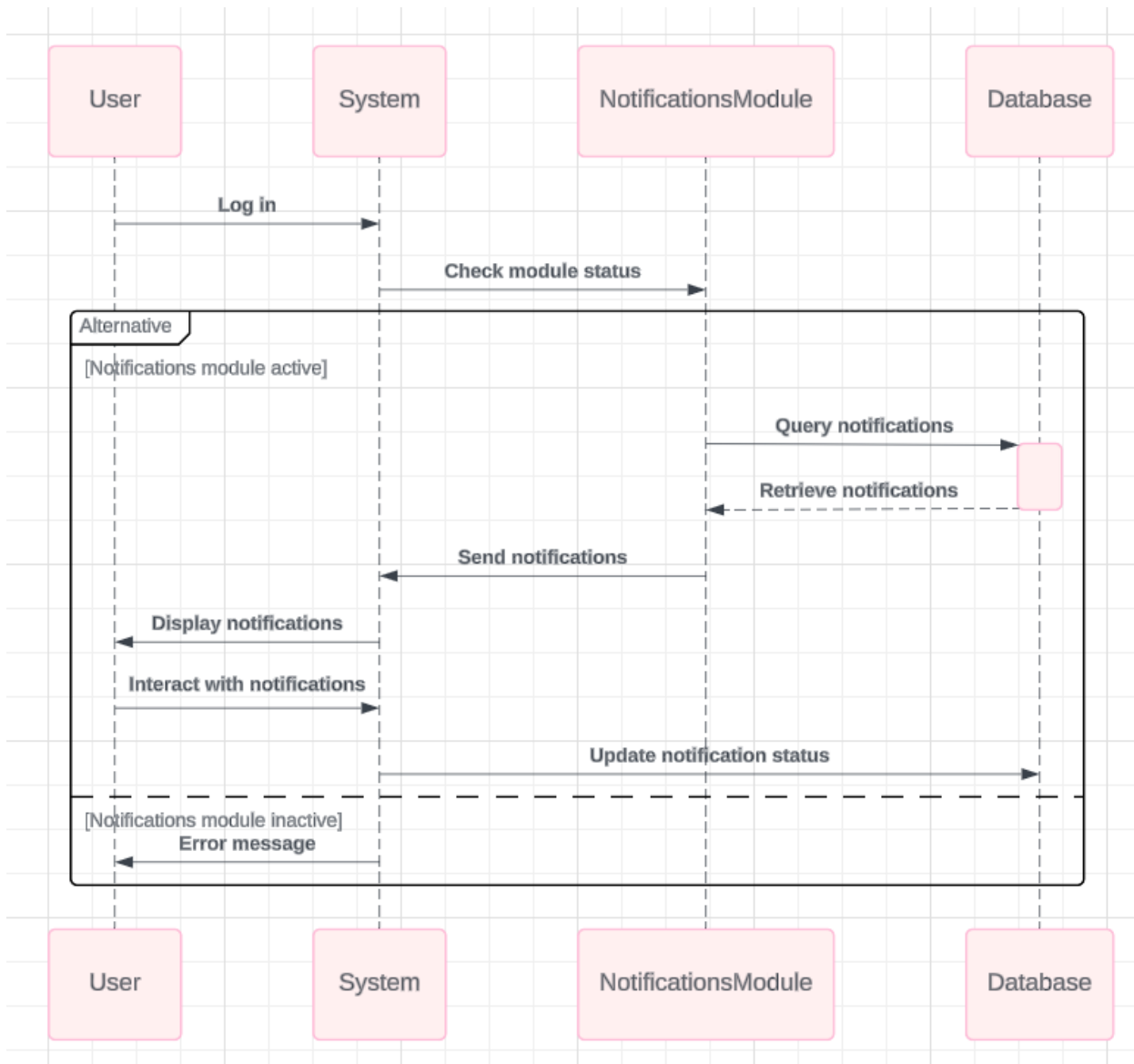
12.6 Create Personalized Plans



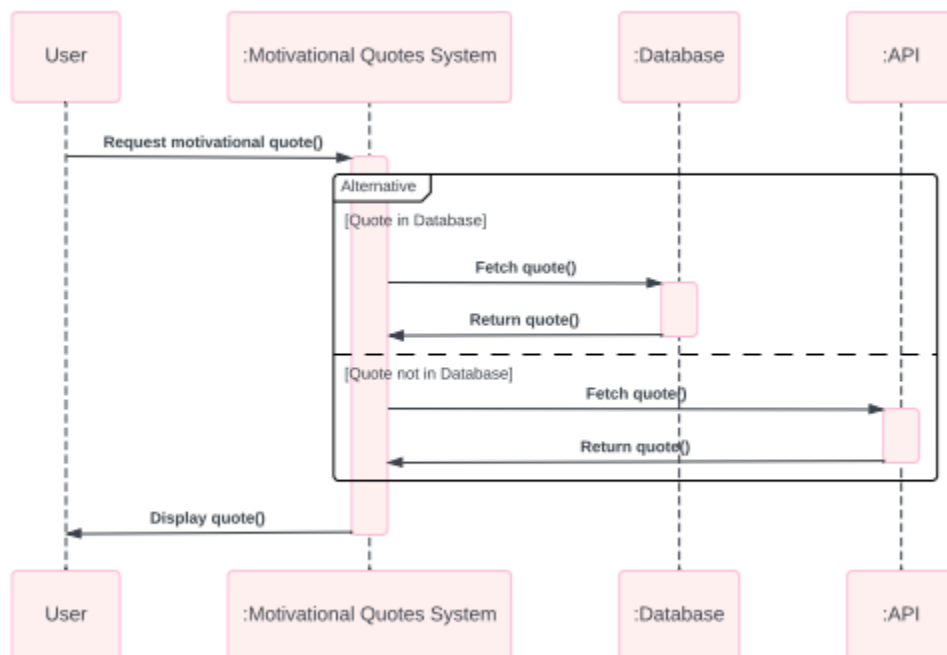
12.7 Track Progress Report



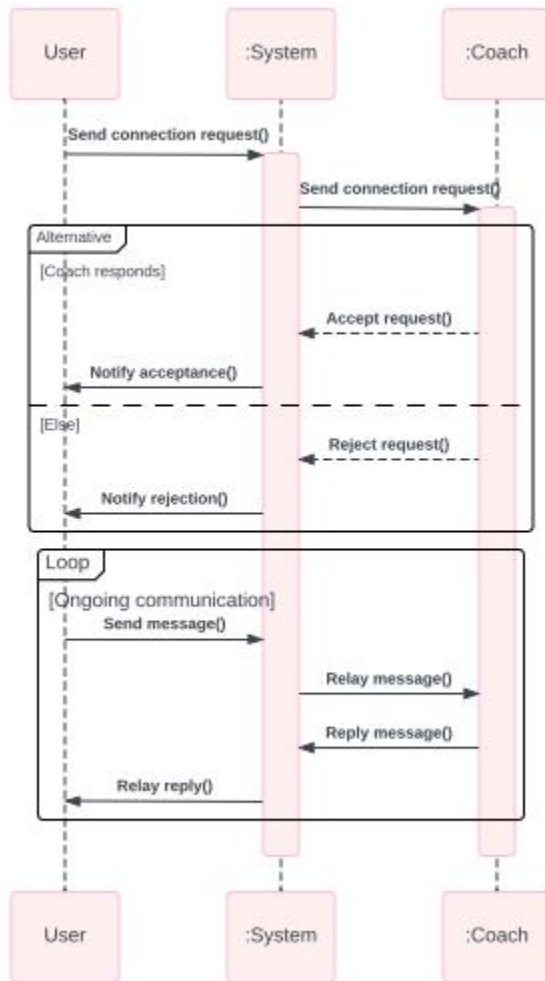
12.8 Notifications



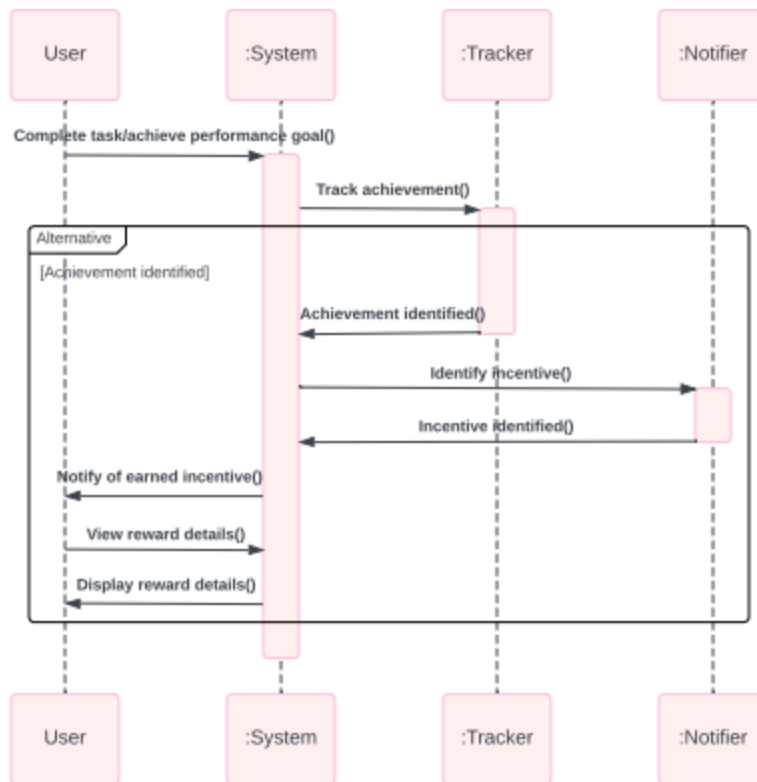
12.9 Get Motivational Quotes



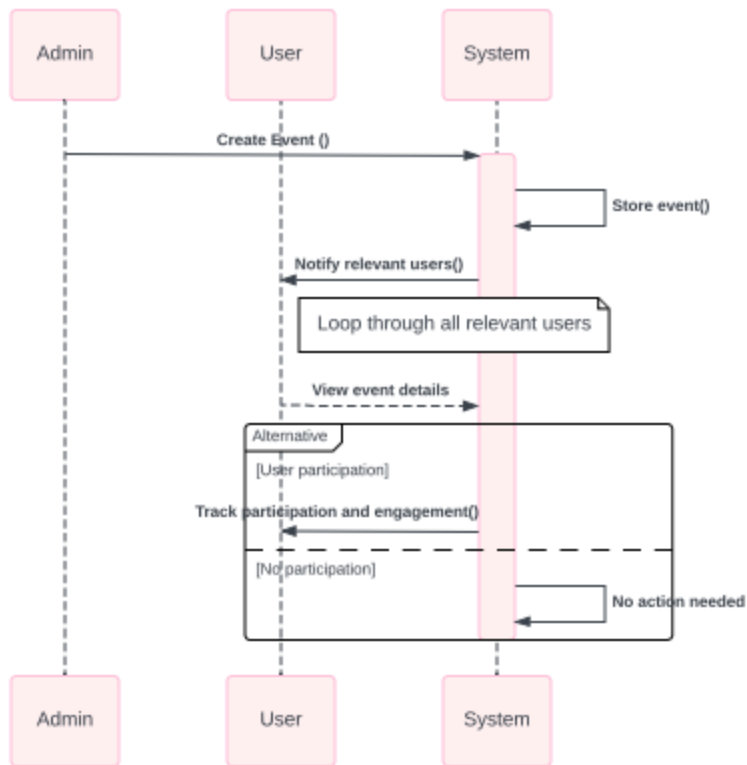
12.10 Connect With Coach



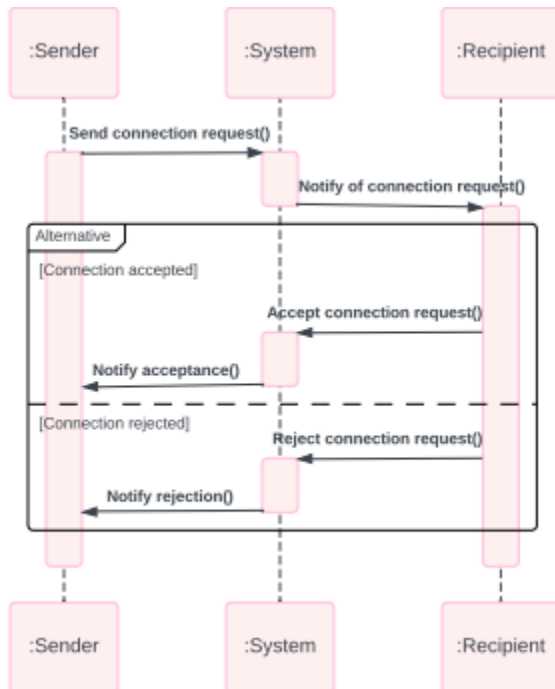
12.11 Provide Incentives



12.12 Create Events



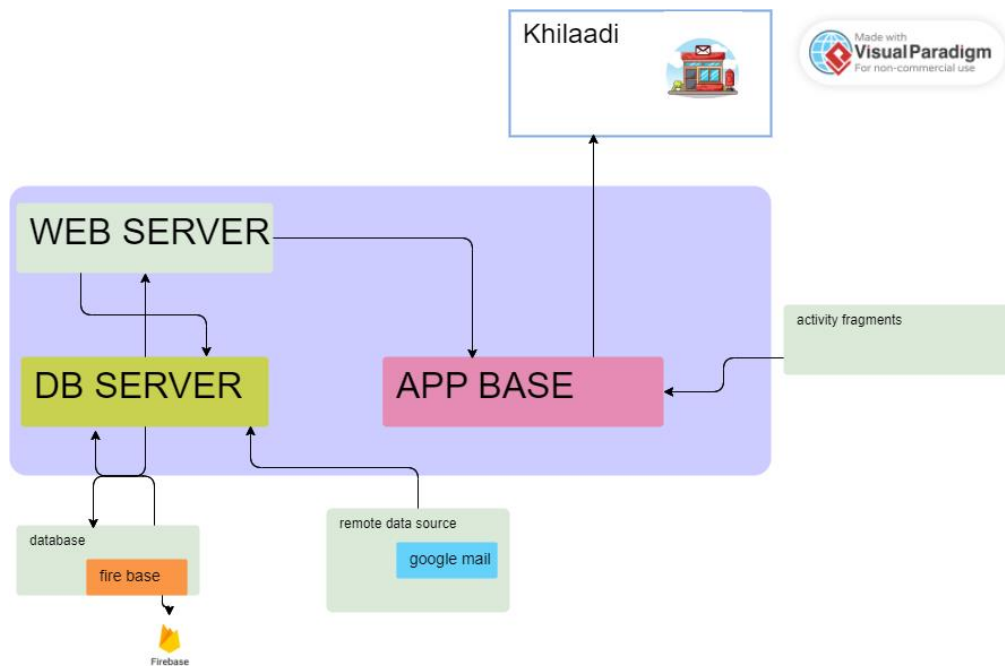
12.13 Connect with other users



13. ER DIAGRAM



14. Architecture Diagram



15. Implementation Details

15.1 Front-End Implementation

As for front-end development, we use Kotlin. Our concentration has been geared towards excellent graphic design of the application thus we have concentrated on efficiency in speed, usability and sleek nature. This website's user interface also uses new materials for design and is designed in a responsive manner.

15.2. Back-End Implementation

For back-end development, we are using Google Firebase for our required back-end development. The database and back-end features are implemented as follows:

- We collect user information during the registration process.

- Firebase Authentication is used to allow users credentials to log in through email and password or phone number. Every user has a sub-table based on their avatar and characters attributes.
- A user also has a sub-table of created and attended meetings with all the details regarding those meetings.
- The data stored in the different tables have been described in the ER diagram that is why all of them are relevant.

Why Firebase?

We chose Firebase for the following reasons:

Real-Time Database: Firebase provides for real time data sharing which is crucial in the case of many users.

Server Management: Firebase provides all the backend support, right from security and the like, so as to free the developers.

Easy Integration: There are several SDKs and APIs available in Firebase that can be easily implemented in the Kotlin-based application.

Authentication and Security: Firebase comes with authentication services to help deal with user credentials and ensuring security of users information.

15.3. Model Training as well as Integration

We use Python in model training hence integrating TensorFlow Lite and PyTorch allowing us to build and deploy systems such as recommender systems and analytics.

TensorFlow Lite: To improve the efficiency of the machine learning models, we employ TensorFlow Lite in the deployment of models across portable devices. This enables real time prediction and low latency out of necessity for on-device artificial intelligence applications.

PyTorch: Dynamic computation graphs are necessary for building and training more complex models which is why PyTorch is employed. They are incorporated into the app to make the AI capabilities better such as the functions for making user-specific predictions and analysis.

After training the models they are taken to the back end, firebase works as a mediator between the kotlin based front end and python based back end. Firebase provides real-time data management, and this will facilitate interaction between the models and the app.

16. Research:

16.1 Summary of article1:

This article introduces and consider the solution to human pose estimation. This is a computer vision problem that may be the core component of tracking human motion and providing objective feedback for physical activities. Nevertheless, estimation of human poses is very difficult owing to the diversity of the postures, masking of body parts and collecting a great number of images.

16.1.2 Critical Analysis

16.1.2.1 Strength

- Can be applied to determine and provide a critique of how a person performs from the onset of a chore to the last stage
- Has room for enhancement for better planning of fitness application

16.1.2.2 Weaknesses

- This can be difficult because of the variety of poses and their occlusions as well as the need of having large data base

16.1.3 Proposed Approach:

Our proposed approach is similar because it employs human pose estimation in examining human movement. Whereas, our approach is aimed at investigation of cricket bowling videos, the research item has a broader perspective and is focused on the videos of any exercises.

Our approach has potential benefits to cricketers by allowing them to be able to self- assess their bowling skills when given constructive feedback. This feedback leads to improvement in the precision and reproducibility of the players.

16.2 Summary of article2:

Deep Neural Networks and pose estimation techniques are employed in this research work to evaluate left arm spin bowling in cricket. Its goal is to build a unique device for real-time techniques analysis with use of Convolutional Neural Networks thus allowing players to correct their mistakes personally rather than needing a coach present to guide them. The project has some challenges regarding dataset acquisition and is currently limited by high noise videos and multi action videos. Other areas of development entail extending the current application to other types of players and different sports as well and developing headroom for advanced situations. This new approach is a technological breakthrough in the analysis of cricket amongst other sports.

16.2.2 Critical Analysis

16.2.2.1 Strength

- **Innovation:** The technology embeds advanced DNNs and pose estimation approaches to a specific niche area of sports analysis (left-arm spin bowling in cricket) that is still in the nascent stages of development.
- **Self-improvement tool:** It allows players to assess and enhance their skills even in situations when professional coaching is not available, which is highly useful.
- **Comprehensive analysis:** The integration of pose estimation techniques with CNNs creates room for the in-depth, detailed and multi-aspect analysis of the studied techniques of bowling.

16.2.2.2 Weaknesses

- **Limited dataset:** There is a limited availability of existing literature and data on left-arm spin bowling techniques that can affect the performance of the system to some extent and particularly in the early stages.
- **Video Quality Dependency:** The given version can be deficient in handling any footage of poor resolution making it useful to people with high-definition recording gadgets only.
- **Single player focus:** Incorporating multi-player video footage, which in many cases will be important especially for team training or during matches, is hard because of the inability to use a single video to target different players.

16.2.3 Proposed Approach:

- The problem statement defined reflects the research motivation outlined in section 1.4. Both understand there is a difference between theory and practice in cricket specially when it comes to spin bowling.
- Our approach relies on AI and machine learning, and this is in tandem with another study referred to earlier in the work “State-of-the-Art Deep Learning in Athletic Performance Analysis”. The above explains why your project of analyzing sports performance should fit in the present trends of analysis and evaluation.
- The plan proposed for analyzing bowling videos is well-supported by several references cited in the current literature search, including Video Analysis of Cricket Bowling Actions for Biomechanics and Performance Evaluation, as well as the Impact of Video-Based Training on Cricket Bowling Accuracy and Kinematics.
- Our goal to give them feedback instantly with certain actions tied up to the researched problem stated in section 1.3 as the benefits of the suggested real time video analysis system for the players and the coach.
- The choice made to work on an Android application is connected with the study “Towards Real-Time Sports Motion Analysis on Mobile Device” which indicates that current technological trends in sports analysis are being followed in the present work.

- Specifically, this objective of ours to find technical issues and be suggested correlates with the general research motivation of enhancing bowling approaches with technology aid.
- Paying particular attention to the presentation of the analysis meant for amateur and semi-professional players, our respond to the limited bodily coaching in effect in world regions such as Pakistan. Our research motivation generally addresses the interplay between the theoretical and practical application of the information.
- The ideas of tracking the progress over the time correspond to the approach of analyzing and developing the technique that was described in the research motivation part.

16.3 Summary of article3:

Thus, the methodology for predicting cricket strokes is presented in the paper “Enhancing Cricket Performance Analysis with Human Pose Estimation and Machine Learning”. The authors concentrate on eight types of movement in playing cricket, the cover drive, pull, flick and so on and use Media Pipe for processing movement data from video data which includes human body key points. Strokes recognition is completed with several machine learning algorithms: Random Forest (RF), Support Vector Machines (SVM), k-Nearest Neighbors (KNN), and Long Short-Term Memory (LSTM) networks; RF has 99.77% accuracy. Accordingly, this study targets actual application of computer vision and ML in opportunity assessment which might be developmental to players, especially when it comes to stroke prediction, although the study will be universal with all kinds of players.

16.3.2 Critical Analysis

16.3.2.1 Strength

- **High Accuracy:** Another strength realized from the study is the use of the RF model with 99.77% accuracy, an indication for a sound model for stroke classification.
- **Comprehensive Dataset:** To get a large and diverse dataset the authors recorded video from multiple sites to capture all kinds of strokes.
- **Innovative Approach:** The integration with computer vision with human pose estimation along with ML models is perfect now for analyzing performance especially in games like cricket.
- **Real-Time Application:** Describing how such a system can be used in real-time coaching, the study clinically and practically demonstrates how this research can be used to provide instant feedback to the cricketers.

16.3.2.2 Weaknesses:

- **Limited Scope on Bowling:** Though the study is precise with regard to the prediction of stroke, it lacks other fundamental dimensions of cricket like bowling assessment. The models and techniques applied are more of batting.
- **Complexity of System Implementation:** The essential mathematics make the approach computationally intensive and video processing demands will also be high hence the approach may not be easily accessible to an amateur player or team with limited computational capital.
- **Generalization:** The study targets eight specific strokes to which it might be restricted by these categories. There are definitely other techniques and strokes that the model might not be able to identify if the training component was not programmed that way.

16.3.4 Proposed Approach:

Your project of contextual-band AI feedback for cricket bowling videos is grounded on the similar ideas as the work of using video data analysis and machine learning. Nevertheless, it pertains only to batting strokes and your project is aimed in assessing bowling performance. Your focus on bowling is complementary to both approaches – the video-based human motion analysis using computer vision, but it is an entirely different idea. In fact, the very same principles of human pose estimation (for example using Media Pipe), or other machine learning models like RF or SVM could be employed to estimate bowling form, trajectory, and any other important parameter in this context.

The goal of your project of generating feedback for semi-professional players in line with the practical application discussed in this paper that involves offering coaching points in real-time or nearly real-time basis. You have the similar chances of using key point extracting and machine learning classifier to analyze errors in bowling, which then can be personalized and used for the skill progression. In the best case, the paper can help you to use similar technologies in your project's sphere.”

17. Conclusion:

Our goal is to ensure that no talent goes unnoticed and that every child in Pakistan receives the guidance they need to enhance their skills and realize their full potential. We are committed to nurturing young talent and helping them become stars in their own light.